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RESEARCH

Pareto-Based Joint Optimization for Rapid Transit Lines: Integrating Flexible Routing, Stop Planning, and Train Scheduling under Time-Dependent Demand

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Abstract
Rapid transit systems (RTSs) are essential for sustainable urban mobility, especially in high-density metropolitan areas where passenger demand varies significantly over time and space. Efficient planning of such systems requires balancing service quality with operational cost under complex operational constraints. This study presents a multi-objective framework for the joint optimization of frequency setting, stop and path planning, timetabling, and train circulation in RTSs under time-dependent demand. A mixed-integer nonlinear programming (MINLP) model is formulated to capture these interdependent planning decisions while considering first-in-first-out-based traffic assignment and operational limitations. To solve this large-scale problem, a tailored NSGA-II algorithm is developed, enabling flexible control over journey patterns, routing strategies, and rolling-stock allocation. The framework is validated via extensive numerical experiments based on real-world data from Xi'an Metro Line 1. Compared with traditional operations such as all-stop, skip-stop, and short-turn strategies, the proposed approach achieves reductions in operational costs and passenger waiting times while guaranteeing full coverage of passenger demand. The results highlight the potential of demand-responsive planning for improving service efficiency and resource use in urban transit systems.

Keywords Frequency setting · Stop planning · Path planning · Train timetabling · Train circulation planning · Traffic assignment · Pareto optimization

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1 Introduction

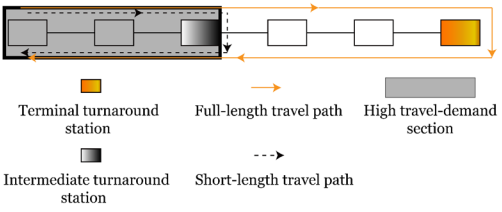
1.1 Background and motivation

Rapid transit systems (RTSs) are a cornerstone of sustainable urban transportation, offering high-capacity mobility with low environmental impact. In recent years, urbanization and increasing commuting needs have led to a surge in RTS ridership. However, this growth poses operational challenges such as platform congestion, longer waiting times, and inefficient resource usage, especially during peak hours.

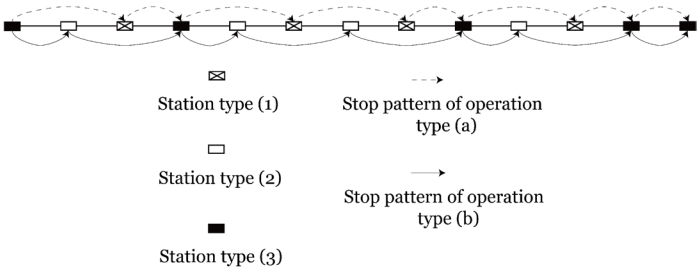
To address these issues, operators are expected to develop optimised daily train plans that balance passenger comfort and operational efficiency. Planners often deploy targeted strategies like short-turning and skip-stop operations to adapt services dynamically to spatial and temporal demand variations. Figure 1a and b illustrate these travel patterns, highlighting their potential to enhance capacity utilisation in high-demand segments.

1.2 Challenges in Multi-stage Planning

RTS operation planning involves multiple interdependent stages: line and stop planning at the strategic level, timetabling and rolling stock allocation at the tactical level, and real-time rescheduling at the operational level, as depicted in Fig. 2. Traditionally, these stages are optimised sequentially to reduce computational complexity. This decomposition, however, overlooks the inherent interdependencies among planning decisions, often resulting in infeasible or suboptimal solutions.



(a) Short-turn and full-length operating patterns.



(b) Station types and skip-stop patterns.

Fig. 1 Illustration of journey patterns and station types used in the model

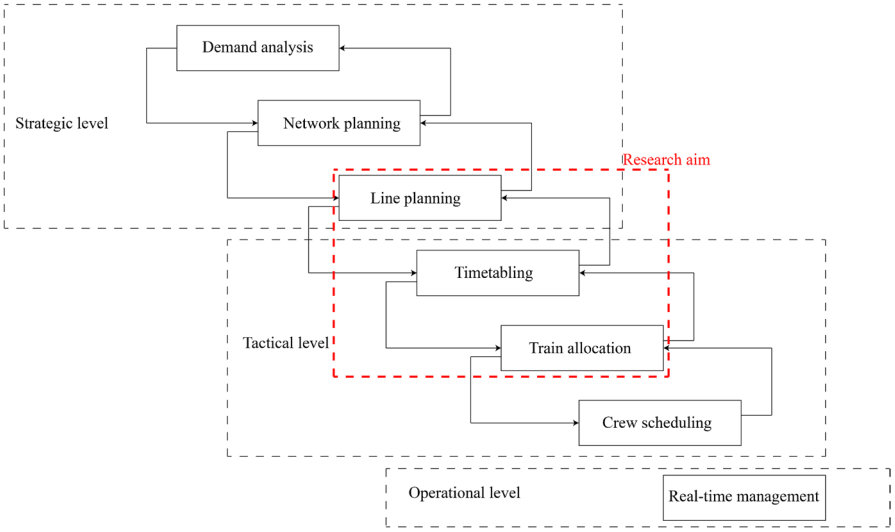


Fig. 2 Chronological stages of rapid transit planning

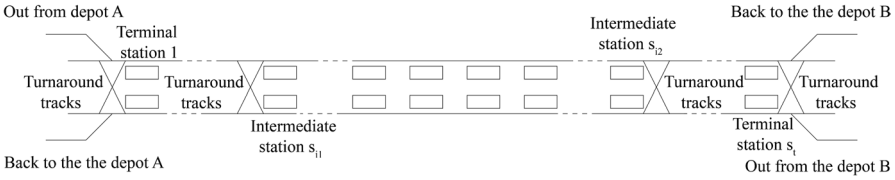


Fig. 3 Layout of the RTS considered in the current study

For instance, studies on timetabling by Cao and Feng (2020); Jamili et al. (2012); Yalçınkaya and Bayhan (2012); Zhang et al. (2024) have shown that optimising train schedules without jointly considering rolling stock circulation can lead to inefficient train usage. To address this, recent works have proposed frameworks that combine multiple planning stages. For example, Burggraeve et al. (2017); Schöbel (2017); Yue et al. (2017) introduce models that integrate line planning, scheduling, and vehicle circulation, demonstrating that joint optimization improves both service quality and operational cost-effectiveness.

Despite their advantages, these integrated approaches often require complex mathematical formulations that compel researchers to restrict decision variables to maintain tractability. Consequently, the flexibility of planning (e.g., flexible skip-stop or partial line coverage) is often limited, hindering real-world applicability (Fig. 3).

1.3 Objectives and Contributions

In response to the limitations of sequential and partially joint RTS planning, this study develops a unified multi-objective framework that captures the strong interdependencies among frequency setting, stopping patterns, routing strategies, timetabling, and

rolling-stock circulation under time-varying demand. The model incorporates short-turning and skip-stop strategies, enabling demand-responsive service patterns that improve both efficiency and user experience.

The passenger-assignment component updates train occupancy along each journey according to boarding and alighting flows. This allows the model to evaluate service feasibility under time-dependent origin–destination demand while avoiding an unnecessary emphasis on capacity constraints as a separate source of novelty.

The main contributions of this paper are fourfold:

1. **Time-dependent MINLP for joint RTS planning.** We formulate a mixed-integer nonlinear programming model that jointly optimises frequency setting, routing decisions (full-length vs. short-turn), stopping patterns, train timetables, and train circulation under time-dependent passenger demand. Passenger-loading constraints link passenger assignment and rolling-stock use, so that the available capacity on each train is updated according to cumulative boarding and alighting decisions along its path.
2. **Demand-responsive journey design.** The modeling framework explicitly accommodates both short-turning and skip-stop strategies within a single, coherent structure. By allowing different journey patterns to coexist and adapt to spatial and temporal demand variations, the model captures a much richer design space than traditional all-stop or fixed-pattern schemes, while preserving operational feasibility.
3. **Tailored NSGA-II with feasibility-preserving encoding.** We develop a problem-specific NSGA-II procedure in which chromosomes encode journey selection, routing, and stopping decisions, whereas timetable generation, train circulation, and passenger assignment are embedded in the evaluation step. Feasibility is enforced by construction through operators that respect headway, turnaround, and capacity constraints, rather than relying on generic penalty functions or ad hoc repair rules. The MINLP formulation thus acts as a structural backbone for the metaheuristic.
4. **Comprehensive numerical evaluation on a real metro line.** Using smart-card demand data from Xi'an Metro Line 1, we conduct extensive numerical experiments. The proposed approach is benchmarked against single-objective and multi-objective metaheuristics, and against traditional all-stop, skip-stop, and short-turn patterns. We analyze convergence behavior, Pareto-front structure, and parameter sensitivity, and derive full-day timetables and circulation plans for realistic operating scenarios, highlighting the operational implications for transit agencies.

The remainder of this paper is organized as follows. Section 2 reviews the related literature and highlights research gaps. Section 3 introduces the problem description and operating assumptions. Section 4 presents the mathematical formulation. Section 5 describes the proposed NSGA-II-based solution method. Section 6 reports the numerical experiments, and Section 7 concludes the paper.

2 Literature Review

Efficient operation planning of RTSs has been extensively studied across multiple levels of decision-making: strategic, tactical, and operational. This section reviews key contributions in the field, organised along three analytical axes: (i) multi-stage optimization, (ii) path and stop planning, and (iii) remaining gaps in flexibility and scalability.

2.1 Multi-stage Optimization in RTS Planning

Traditionally, RTS operation planning has followed a stage-wise process. However, sequential optimization often leads to inefficiencies by neglecting interdependencies across planning levels. To improve coordination, various studies have proposed frameworks targeting tactical and strategic levels.

Initial efforts such as those of Van Lieshout (2021); Zhou et al. (2022, 2023) combine timetabling and rolling stock assignment, focusing on energy savings and resource circulation. More recent works by Schettini et al. (2022); Xue et al. (2022); Zhu et al. (2023) address joint line planning and timetabling under demand-driven or robust strategies, while Wang et al. (2021) incorporates energy-aware scheduling under automatic train operation scenarios.

A smaller group of studies extends integration to three levels: line planning, timetabling, and vehicle circulation. For instance, Mo et al. (2021); Zhao et al. (2021) propose exact MILP formulations, while Zhou and Oldache (2021) uses a MINLP-based multi-objective framework. These models show strong performance in simulation or real-world environments, but often impose structural simplifications or rely on commercial solvers. Studies such as Hu et al. (2023); Yuan et al. (2022) further embed short-turning and skip-stop strategies into circulation planning, balancing service quality with operational robustness.

However, some advanced models depend on weight factor adjustments or fixed scenarios (Dong et al. 2020; Wu et al. 2021; Zhou et al. 2022), which may limit their adaptability. Others, such as Parbo et al. (2018); Schöbel (2017); Shi et al. (2017); Wang et al. (2017), propose heuristics or deterministic algorithms tailored to narrow settings, reducing their generalisability.

2.2 Path and Stop Planning

Designing train routes (paths) and station stopping patterns constitutes a core component of rapid transit service planning, particularly under uneven demand. Path flexibility is often achieved via short-turning (Ding et al. 2018; Hu 2019; Zhou et al. 2022), whereas stop flexibility includes skip-stop, express, or zonal strategies (Hu et al. 2023; Wu et al. 2021; Yang and Du 2021; Zhang et al. 2024; Zhao et al. 2021).

Several studies link these strategies with timetabling or train allocation. For example, Blanco et al. (2020); Ghaemi et al. (2018); Schettini et al. (2022) tie path planning to tactical-level operations, including timetabling. Others such as Yuan et al. (2022); Zhang et al. (2017, 2018) combine path planning with both timetabling and train circulation optimization. Stop planning has been modeled with timetabling in

works such as Hu et al. (2023); Jamili and Aghaee (2015); Parbo et al. (2018); Wu et al. (2021), while Zhao et al. (2021) incorporates stop planning with both timetabling and train circulation planning.

Only a few studies, namely (Feng et al. 2020; Ulusoy et al. 2010; Zhu and Goverde 2019), address both stop and path planning simultaneously. Among them, Zhu and Goverde (2019) includes rescheduling under disruption scenarios, yet still depends on a limited set of predefined journey types. Similarly, Dong et al. (2020); Wang et al. (2017) propose advanced skip-stop formulations, but do not explore adaptive route adjustments. These approaches often lack sensitivity analysis or scalability mechanisms.

2.3 Summary of Comparative Insights

Table 1 summarizes representative contributions on rapid transit planning, highlighting the decision elements jointly considered in each model: frequency setting, stop planning, path planning / short-turning, timetabling, train circulation, and time-dependent travel demands. The table focuses on studies that are either conceptually closest to our work or frequently cited in RTS planning.

Table 1 Comparison of representative rapid transit planning studies

Study	Decision elements						Modeling & Solution
	Freq.	Stop	Path	Timet.	Circ.	TD demand	
Van Lieshout (2021)	–	–	–	✓	✓	✓	MILP; Exact solution
Zhou et al. (2023)	–	–	–	✓	✓	✓	MINLP; Particle swarm algorithm
Zhou et al. (2022)	–	–	–	✓	✓	–	MINLP; Linearization & Tabu search
Schettini et al. (2022)	–	–	✓	✓	–	✓	MILP; Exact solution
Xue et al. (2022)	–	–	–	✓	✓	✓	INLP; Tabu search algorithm
Zhou et al. (2025)	✓	✓	–	✓	✓	✓	MINLP; Two-stage heuristic algorithm
Mo et al. (2021)	✓	–	–	✓	✓	✓	MINLP; Dynamic programming algorithm
Zhao et al. (2021)	–	✓	–	✓	✓	✓	MINLP; Iterative searching algorithm
Zhou and Oldache (2021)	✓	–	–	✓	✓	–	MINLP; Simulated annealing algorithm
Zhu et al. (2023)	–	–	✓	✓	–	✓	MINLP; Linearization & Exact solution
Yuan et al. (2022)	✓	–	✓	✓	✓	–	MINLP; Genetic algorithm
Dong et al. (2020)	✓	✓	–	✓	–	✓	MINLP; ALNS algorithm
Feng et al. (2020)	✓	✓	✓	–	–	–	MILP; Exact solution
Zhu and Goverde (2019)	–	✓	✓	✓	✓	–	MILP; Exact solution
This paper	✓	✓	✓	✓	✓	✓	**MINLP; Tailored NSGA-II.**

Most existing contributions integrate at most two or three tactical elements. For example, Van Lieshout (2021); Zhou et al. (2022), and Zhou et al. (2023) develop integrated timetabling and rolling-stock circulation models, sometimes coupled with frequency-setting decisions, but with limited flexibility in stop or route design. Other works, such as Schettini et al. (2022); Xue et al. (2022); Zhu et al. (2023), jointly address line planning and timetabling under demand-driven or robust frameworks, while still relying on predefined stopping patterns and static route structures.

A second group of studies extends the level of integration to three dimensions, typically combining line planning, timetabling, and circulation, as in Mo et al. (2021); Zhao et al. (2021); Zhou and Oldache (2021). Although these models are structurally rich, they usually operate under static or aggregated demand and maintain rigid all-stop or pre-specified patterns. Short-turning and skip-stop are explicitly embedded in only a few studies, such as Hu et al. (2023); Yuan et al. (2022); Dong et al. (2020); Feng et al. (2020); Zhu and Goverde (2019), where they are either restricted to fixed templates or analyzed in isolation from circulation decisions.

The last row of Table 1 positions the present study relative to this body of work. In contrast to prior models, our MINLP formulation simultaneously: (i) sets frequencies, (ii) designs flexible stop and path patterns (including short-turn and skip-stop), (iii) generates conflict-free timetables, (iv) plans train circulation, and (v) evaluates these decisions under time-dependent demand via FIFO-based passenger assignment. This level of joint treatment of frequency, stop/path flexibility, timetabling, and rolling stock under realistic, time-varying demand has, to the best of our knowledge, not been reported in the existing literature.

3 Problem Statement

This study considers a bi-directional rapid transit line composed of multiple stations and equipped with turnaround facilities at both terminal and selected intermediate stations. The system supports flexible operations, including short-turning and station-skipping strategies, in response to time-varying passenger demand.

3.1 Rail Line Structure and Operation Zones

The transit line is defined as a double-track system linking a set of s_t stations. The model allows depot access at terminal stations, while the active depot configuration depends on the case-study setting. In the numerical experiments, the small-scale instance uses two terminal depots, whereas the full-day instance uses a single depot connected to station 1. Intermediate turnaround options are available at stations s_{i1} and s_{i2}. The upstream direction flows from station 1 to station s_t, while the downway direction flows in reverse. Turnaround operations are allowed at both ends and designated intermediate stations, enabling partial-length services.

3.2 Journey Configuration and Scheduling Horizon

The operational day is divided into discrete time intervals, indexed by u, within global limits t_{earliest} and t_{latest}. Four classes of potential journeys are defined: full-length journeys in up-way and down-way directions, and short-turn journeys originating or terminating at intermediate stations s_{i1} and s_{i2}. Each journey class is represented by a candidate set J^{up} for those operating in the up-way direction and J^{dn} for those operating in the down-way direction, with spacing between consecutive journeys controlled via a tunable time unit u, affecting both model scale and complexity. Figure 4 shows the sets of potential journeys in a single time interval.

3.3 Operational Constraints and Assumptions

To ensure feasibility and consistency with typical RTS infrastructure, the following assumptions are adopted.

Assumption 1 (Single-train station occupancy). At any given time, at most one train per direction may occupy a given platform. This reflects standard safety regulations and infrastructure limitations, particularly in single-platform or narrow-station layouts, and ensures conflict-free sequencing of arrivals and departures.

Assumption 2 (No overtaking). The network does not contain overtaking tracks. Once a train enters a track segment, it cannot be bypassed by another train on the same line. This assumption restricts timetabling flexibility and requires careful coordination of journey sequencing to avoid bottlenecks and cascading delays.

Assumption 3 (Fixed train composition and capacity). All trains operate with a fixed formation and passenger capacity throughout the day. This reflects common metro operations, where dynamically adjusting train length is logistically infeasible, and ensures consistency in rolling-stock performance and scheduling constraints.

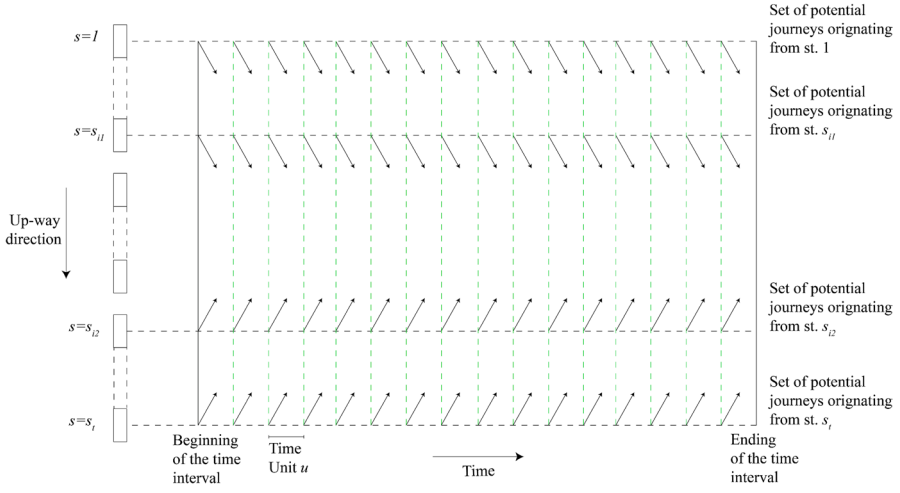


Fig. 4 The principle of potential journeys

Assumption 4 (Fully informed passenger behavior). Passengers are assumed to have full knowledge of the operating schedule and to choose their boarding and alighting stations accordingly. This enables the use of deterministic FIFO-based traffic assignment, simplifying the estimation of demand satisfaction, waiting times, and boarding events at each station.

These assumptions support the tractability of the multi-objective formulation while maintaining sufficient realism for practical application. They also ensure that the scheduling algorithm incorporates safety margins and capacity-aware constraints without requiring stochastic simulation of passenger behavior.

4 Mathematical Formulation

The optimization model aims to jointly determine which journeys to operate, their path and stop plans, departure and arrival schedules, and corresponding train assignments, while accurately modeling passenger flows and train capacities. The goal is to balance passenger satisfaction and operational efficiency under strict infrastructure and resource constraints.

4.1 Decision Variables

The optimization model relies on five interrelated groups of decision variables. First, journey selection is captured by a binary variable I_j^s, which equals 1 if journey j is scheduled within the planning horizon and 0 otherwise. Second, routing and stop planning are modelled through binary variables P_j^s and S_j^s: the former indicates whether journey j operates a full-length or short-turn path, while the latter indicates whether journey j stops at station s.

Third, timetabling decisions are represented by continuous variables $E_j^{\mathrm{dep},s}$ and $E_j^{\mathrm{arr},s}$, denoting departure and arrival times of journey j at station s, subject to dwell and running-time constraints. Fourth, train assignment and circulation are encoded through binary variables O_j^s, R_j^s, and $L_{j,j'}^s$. Figure 5 shows some applications of the decision variables discussed in this section. Finally, passenger flow assignment is described by integer variables that track the movements of passenger flows along the scheduled journeys; these are detailed in Section 4.3.

A full summary of all decision and intermediate variables is provided in Table 2. We summarise the input parameters involved in the model in Table 3.

4.2 Model Constraints

The constraints of the model are presented below as a single unified set. They enforce safety headways, consistency of path and stopping decisions, timetabling feasibility, synchronisation of linked journeys, turnaround operations, and daily operational time limits.

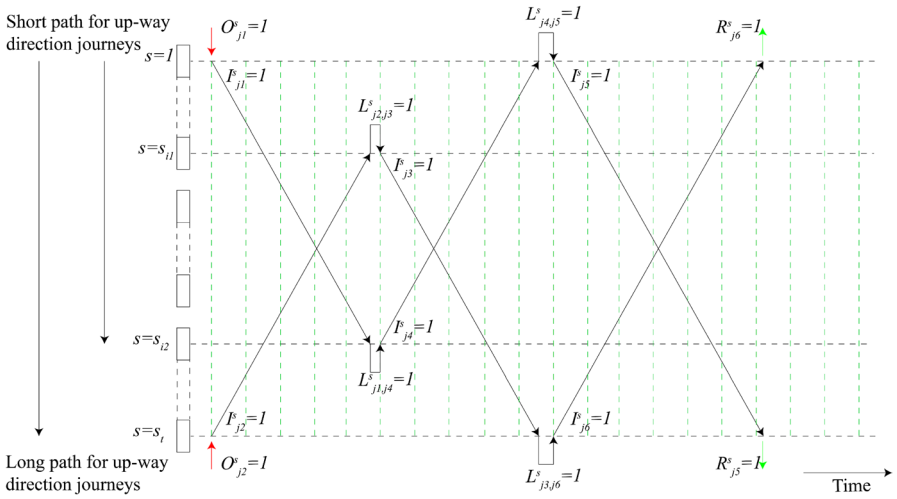


Fig. 5 Applications of journey selection and train allocation variables

For safety, the time difference between two consecutive journeys must be at least the minimum headway $t_{\min}$. For service regularity, it must not exceed the maximum headway $t_{\max}$. These conditions apply only to pairs of *consecutive* journeys (j, j'), defined as those satisfying $\sum_{j''=j+1}^{j'-1} I^s_{j''} = 0$:

$$(j' - j) \cdot u \;\geq\; t_{\min} \cdot I^s_j \cdot I^s_{j'}, \quad \forall s \in S,\ \forall (j, j') \in J^{\mathrm{up}} \cup J^{\mathrm{dn}},\ j' > j, \tag{1}$$

$$(j' - j) \cdot u \;\leq\; t_{\max} \cdot I^s_j \cdot I^s_{j'}, \quad \forall s \in S,\ \forall (j, j') \in J^{\mathrm{up}} \cup J^{\mathrm{dn}},\ j' > j. \tag{2}$$

Here, u is the time-unit parameter.

A journey can only be assigned a path if it is scheduled ($I^s_j = 1$):

$$P^s_j \;\leq\; I^s_j, \qquad \forall s \in S,\ \forall j \in J^{\mathrm{up}} \cup J^{\mathrm{dn}}. \tag{3}$$

A stop event ($S^s_j = 1$) or skip event ($S^s_j = 0$) can only be assigned when the journey is active ($I^s_j = 1$). Within the short-turn segment, stops follow active journeys; beyond it, stops are allowed only on full-length paths ($P^s_j = 1$):

$$\begin{aligned} S^s_j \;&\leq\; \sum_{s'} I^{s'}_j, \quad \forall\big((s \leq s_{i2} \wedge j \in J^{\mathrm{up}}) \;\vee\; (s \geq s_{i1} \wedge j \in J^{\mathrm{dn}})\big), \\ S^s_j \;&\leq\; \sum_{s'} P^{s'}_j, \quad \forall\big((s > s_{i2} \wedge j \in J^{\mathrm{up}}) \;\vee\; (s < s_{i1} \wedge j \in J^{\mathrm{dn}})\big). \end{aligned} \tag{4}$$

Table 2 Decision variables used in the mathematical model

Variable	Description
I_j^s	Binary decision variable that determines whether to schedule journey *j* originating from station *s*, where $I_j^s = 1$ if journey *j* is scheduled and $I_j^s = 0$ otherwise.
P_j^s	Binary decision variable that determines whether journey *j* runs on the full-length path or the short path, where $P_j^s = 1$ schedules the journey to run on the full-length path and $P_j^s = 0$ on the short path.
S_j^s	Binary decision variable that determines whether journey *j* stops or skips station *s*, where $S_j^s = 1$ schedules the journey to stop at station *s* and $S_j^s = 0$ to skip station *s*.
$E_j^{\mathrm{dep},s}$	Real variable that gives the departure time of journey *j* from station *s*.
$E_j^{\mathrm{arr},s}$	Real variable that gives the arrival time of journey *j* at station *s*.
O_j^s	Binary variable that decides whether the train allocated to journey *j* comes out of the depot connected to station *s*.
R_j^s	Binary variable that decides whether the train allocated to journey *j* returns to the depot connected to station *s*.
$L_{j,j'}^s$	Binary variable that decides whether the train allocated to journeys *j* and j' turns around at station *s*.
$C_{j,\mathrm{time}}^s$	Integer variable representing the dwell-time-based passenger-throughput capacity of journey *j* at station *s*.
$C_{j,\mathrm{space}}^s$	Integer variable that determines the available capacity in the train allocated to journey *j* when stopping at station *s* based on space constraints.
C_j^s	Integer variable that determines the effective available capacity in the train allocated to journey *j* when stopping at station *s*.
$B_j^{s,s'}$	Integer variable that computes the number of passengers travelling from station *s* to station s' within journey *j*.
B_j^s	Integer variable representing the number of passengers boarding journey *j* at station *s*.
A_j^s	Integer variable representing the number of passengers alighting from journey *j* at station *s*.

Journeys departing from terminal stations $s \in \{1, s_t\}$ have departure times determined by the earliest departure allowed and the index of the journey:

$$E_j^{\mathrm{dep},s} \;=\; I_j^s \cdot \big(t_{\mathrm{earliest}} + (j-1)\cdot u\big), \qquad \forall (s = 1 \wedge j \in J^{\mathrm{up}}) \;\vee\; (s = s_t \wedge j \in J^{\mathrm{dn}}). \quad (5)$$

Arrival times propagate from the previous departure event and depend on the stop/skip decision and the corresponding section running time:

Table 3 Sets and model parameters

Sets	
S	Set of stations.
$S_{\mathrm{i1}}, S_{\mathrm{i2}}$	Sets of intermediate turnaround stations.
S^{dep}	Set of stations with active depot access.
$J^{\mathrm{up}}, J^{\mathrm{dn}}$	Sets of potential journeys in upward and downward directions.
T	Set of discrete time instants.
I	Set of time intervals.
Parameters	
$t_{\min}$	Minimum allowable time difference between two consecutive journeys.
$t_{\max}$	Maximum allowable time difference between two consecutive journeys.
t_{out}	Time required for a train to exit the depot.
t_{in}	Time required for a train to enter the depot.
t_{turn}	Time required for turning a train between opposite-direction journeys.
$Q_s^{s'}(t)$	Passenger demand travelling from station s to station s' at time instant t.
u_c	Unit cost per time unit for operating trains.
p_v	Unit value of passenger time.
Inf	A sufficiently large positive number.
$\mathrm{JT}_s^{\mathrm{stop}}$	Running time from station s to the next station when the train stops.
$\mathrm{JT}_s^{\mathrm{skip}}$	Running time from station s to the next station when the train skips.
DT_s	Dwell (stopping) time of trains at station s.
C	Maximum carrying capacity of a train.
t_{earliest}	Earliest allowable departure time of the day.
t_{latest}	Latest allowable departure time of the day.
u	Time unit used to distinguish consecutive potential journeys.
s_t	Total number of stations in the rail line.

$$
\begin{aligned}
E_j^{\mathrm{arr},s'} &= \sum_{s''} I_j^{s''} \left(E_j^{\mathrm{dep},s} + S_j^{s'} \cdot \mathrm{JT}_{\mathrm{stop}}^{s} + (1 - S_j^{s'}) \cdot \mathrm{JT}_{\mathrm{skip}}^{s} \right), \\
&\qquad \forall \big((s < s_{i2} \wedge j \in J^{\mathrm{up}}) \ \vee \ (s > s_{i1} \wedge j \in J^{\mathrm{dn}}) \big), \\
E_j^{\mathrm{arr},s'} &= \sum_{s''} P_j^{s''} \left(E_j^{\mathrm{dep},s} + S_j^{s'} \cdot \mathrm{JT}_{\mathrm{stop}}^{s} + (1 - S_j^{s'}) \cdot \mathrm{JT}_{\mathrm{skip}}^{s} \right), \\
&\qquad \forall \big((s_{i2} \leq s < s_t \wedge j \in J^{\mathrm{up}}) \ \vee \ (1 < s \leq s_{i1} \wedge j \in J^{\mathrm{dn}}) \big).
\end{aligned} \tag{6}
$$

Here, $\mathrm{JT}_{\mathrm{stop}}^{s}$ represents the journey time from station s to the next station s' when the train is scheduled to make a stop at s'. Likewise, $\mathrm{JT}_{\mathrm{skip}}^{s}$ represents the journey time within the travel section $[s, s']$ when the train is scheduled to skip station s'. Note that in equalities Eqs. 5–6, the departure and arrival events are functions of the journey-selection variables I_j^s, and events occur only when journeys are scheduled.

Departures from intermediate stations depend on arrival events and stopping time:

$$E_j^{\mathrm{dep},s} = \sum_{s'} I_j^{s'} \left(E_j^{\mathrm{arr},s} + S_j^s \cdot \mathrm{DT}^s\right), \qquad \forall (s > 1 \wedge j \in J^{\mathrm{up}}) \ \vee \ (s < s_t \wedge j \in J^{\mathrm{dn}}). \tag{7}$$

Here, DT^s represents the dwell time of trains at station s.

Figure 6 illustrates the four possible configurations that may arise between two consecutive journeys when they operate in the same direction but have different stopping patterns. These cases determine how arrival and departure events propagate across stations and justify the safety constraints formulated in Eqs. 8–9. In all situations, the model must ensure that trains respect the minimum allowed temporal separation regardless of whether a journey stops or skips an intermediate station.

For any pair of consecutive journeys, departure and arrival events must satisfy:

$$\begin{aligned} E_{j'}^{\mathrm{dep},s} &\geq E_j^{\mathrm{dep},s} + t_{\min} - \mathrm{Inf}\left(2 - \sum_{s'} I_{j'}^{s'} - \sum_{s'} I_j^{s'}\right), \quad \forall s \in S, \ \forall (j,j') \in J^{\mathrm{up}} \cup J^{\mathrm{dn}}, \\ E_{j'}^{\mathrm{arr},s} &\geq E_j^{\mathrm{arr},s} + t_{\min} - \mathrm{Inf}\left(2 - \sum_{s'} I_{j'}^{s'} - \sum_{s'} I_j^{s'}\right), \quad \forall s \in S, \ \forall (j,j') \in J^{\mathrm{up}} \cup J^{\mathrm{dn}}. \end{aligned} \tag{8}$$

Arrival after a previous departure must also respect the safety limit:

$$E_{j'}^{\mathrm{arr},s} \geq E_j^{\mathrm{dep},s} + t_{\min} - \mathrm{Inf}\left(2 - \sum_{s'} I_{j'}^{s'} - \sum_{s'} I_j^{s'}\right), \qquad \forall s \in S, \ \forall (j,j') \in J^{\mathrm{up}} \cup J^{\mathrm{dn}}. \tag{9}$$

All movement events must fall within the operating window:

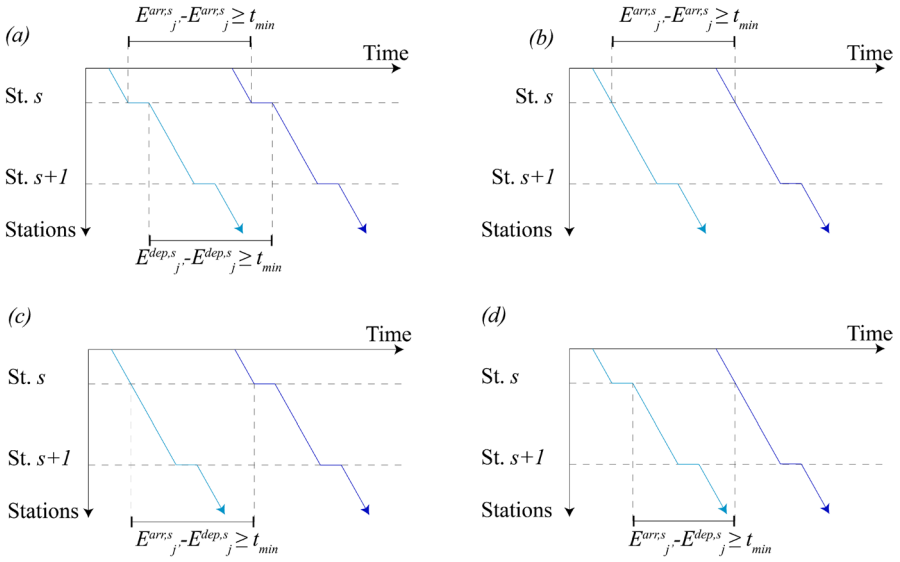


Fig. 6 Possible situations that may arise between two consecutive journeys

$$E_j^{\mathrm{dep},s} \le t_{\mathrm{latest}}, \qquad E_j^{\mathrm{arr},s} \le t_{\mathrm{latest}}, \qquad \forall s \in S,\ \forall j \in J^{\mathrm{up}} \cup J^{\mathrm{dn}}, \tag{10}$$

$$E_j^{\mathrm{dep},s} \ge t_{\mathrm{earliest}}, \qquad \forall s \in S,\ \forall j \in J^{\mathrm{up}} \cup J^{\mathrm{dn}}. \tag{11}$$

At terminal stations, a journey either exits the depot or links to a downstream/upstream journey:

$$\begin{aligned} O_j^s + \sum_{j'} L_{j,j'}^s = I_j^s, \quad \forall s \in S^{\mathrm{dep}},\ \forall j \in J^{\mathrm{up}} \cup J^{\mathrm{dn}}, \\ R_j^s + \sum_{j'} L_{j,j'}^s = I_j^s, \quad \forall s \in S^{\mathrm{dep}},\ \forall j \in J^{\mathrm{up}} \cup J^{\mathrm{dn}}. \end{aligned} \tag{12}$$

At intermediate turnaround stations s_{i1} and s_{i2}, short-turn journeys must reverse direction:

$$L_{j,j'}^s = (1 - \sum_{s'} P_j^{s'}) \cdot I_{j'}^s, \qquad \forall (s = s_{i2},\ j \in J^{\mathrm{up}},\ j' \in J^{\mathrm{dn}}) \ \cup \ (s = s_{i1},\ j \in J^{\mathrm{dn}},\ j' \in J^{\mathrm{up}}). \tag{13}$$

Linked journeys must synchronise arrival and departure events:

$$\begin{aligned} E_{j'}^{\mathrm{dep},s} \ge E_j^{\mathrm{arr},s} + \mathrm{DT}^s \cdot S_j^s + t_{\mathrm{turn}} - \mathrm{Inf}(1 - L_{j,j'}^s), \\ \forall (s \in \{1, s_{i1}\},\ j \in J^{\mathrm{dn}},\ j' \in J^{\mathrm{up}}) \ \vee \ (s \in \{s_{i2}, s_t\},\ j \in J^{\mathrm{up}},\ j' \in J^{\mathrm{dn}}). \end{aligned} \tag{14}$$

Here, t_{turn} represents the minimum turnaround time between two consecutive journeys j and j'.

Figure 7 illustrates the turning operation at an intermediate turnaround station. Journey j_1 arrives from one direction and is linked to journey j_2 departing in the opposite direction after the required turning time. During this interval, the turnaround track is exclusively occupied by the pair (j_1, j_2), so only one journey may arrive at the station. Constraints Eq. 15 enforce this rule by ensuring that only one train can occupy the turnaround section and that all other arrivals occur either after the arrival of j_1 or after the departure of j_2.

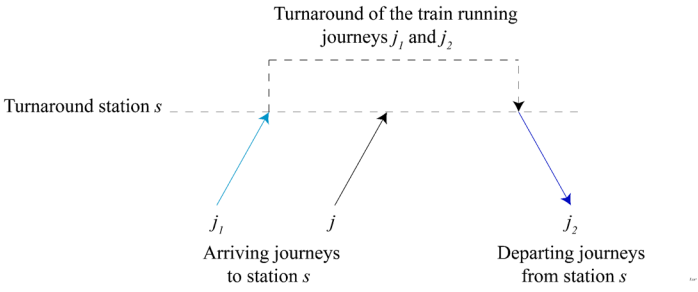


Fig. 7 Arrival limitation of trains during a turnaround operation

$$\begin{aligned} \mathrm{Inf}\cdot(I^{s}_{j_1}-1) \;\le\; \sum_{s'}\sum_{j} I^{s'}_{j} \;\le\; 1+\mathrm{Inf}\cdot(1-I^{s}_{j_1}),\\ \forall\,\big(s\in\{s_{i2},s_t\},(j_1,j)\in J^{\mathrm{up}},j_2\in J^{\mathrm{dn}}\big)\\ \vee\;\big(s\in\{1,s_{i1}\},(j_1,j)\in J^{\mathrm{dn}},j_2\in J^{\mathrm{up}}\big). \end{aligned} \tag{15}$$

Where journey j satisfies:

$$\begin{aligned} E^{\mathrm{arr},s}_{j} &\ge E^{\mathrm{arr},s}_{j_1}L^{s}_{j_1,j_2}+\mathrm{Inf}\big(1-\sum_{s'} I^{s'}_{j}\big),\\ E^{\mathrm{arr},s}_{j} &\ge E^{\mathrm{dep},s}_{j_2}L^{s}_{j_1,j_2}+\mathrm{Inf}\big(1-\sum_{s'} I^{s'}_{j}\big),\\ \forall j\in J^{\mathrm{up}}\cup J^{\mathrm{dn}},&\quad \forall(s,j_1,j_2)\in\Omega^{\mathrm{turn}}, \end{aligned} \tag{16}$$

where

$$\begin{aligned} \Omega^{\mathrm{turn}} = \big\{(s,j_1,j_2): s\in\{s_{i2},s_t\},\; j_1\in J^{\mathrm{up}},\; j_2\in J^{\mathrm{dn}}\big\}\\ \cup\;\big\{(s,j_1,j_2): s\in\{1,s_{i1}\},\; j_1\in J^{\mathrm{dn}},\; j_2\in J^{\mathrm{up}}\big\}. \end{aligned}$$

4.3 Passenger Flow Assignment

To tackle the traffic assignment, we pay close attention to the movements of passengers in and out of each operating journey based on time-dependent travel demands. Travel demand is given by the input parameter $Q^{s'}_{s}(t)$, representing the number of passengers willing to travel from station s to station s' during an instant t. We evaluate flows using the integer variables B^{s}_{j} and A^{s}_{j}, in combination with a passenger-loading capacity variable for each train operating a journey when stopping at a station, considering the maximum carrying capacity of trains C.

When passengers attempt to board a train, the deadline at which they can do so is constrained by time and space: the time limit can be caused by the scheduled departure time of the train from the station, whereas the space limit is reached when the train attains full capacity, as shown in Fig. 8, cases (a) and (b), respectively.

To address situation (a) in Fig. 8, we consider the passenger-throughput limitation associated with the planned dwell time at each station, following the boarding and alighting logic discussed by Zhou and Oldache (2021). For tactical planning, each station is assigned a planned dwell-time budget DT^{s}. Within this planned dwell time, only a finite number of boarding and alighting movements can be completed. Therefore, rather than modelling real-time door-closing behavior, we introduce a dwell-time-based passenger-throughput bound. This bound estimates the maximum number of passengers that can board journey j at station s after accounting for passenger reaction time and the alighting movements that must be completed first. The number

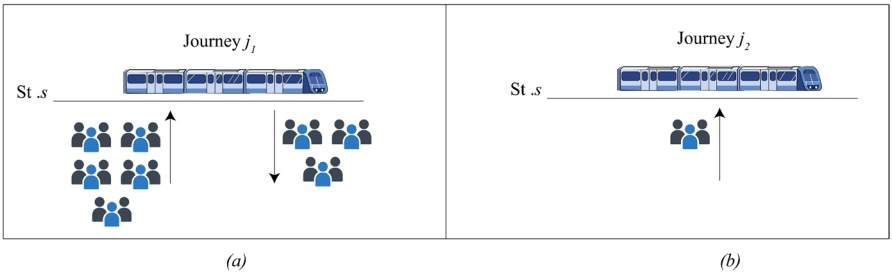


Fig. 8 (**a**) Deadline for passengers to board the train based on time constraints; (**b**) deadline for passengers to board based on space constraints

of remaining passenger movements that can be accommodated within the planned dwell time is computed as

$$C_1 = \left\lfloor \frac{\mathrm{DT}^s - t_r}{t_m} - A_j^s \right\rfloor,$$

where t_r is the passenger reaction time after train arrival and t_m is the average time required for one passenger movement.

$$C^s_{j,\mathrm{time}} = S^s_j \cdot C_1, \qquad \forall j \in J^{\mathrm{up}} \cup J^{\mathrm{dn}},\ \forall s \in S. \tag{17}$$

Constraint Eq. 17 should therefore be interpreted as a planned dwell-time throughput constraint. It does not imply that train doors physically close while passengers are still boarding or alighting. Rather, it limits the number of passengers assigned to a scheduled journey under the planned timetable. In real-time rescheduling applications, this constraint could be replaced by an endogenous dwell-time formulation in which excessive passenger movements extend dwell time and propagate delays to subsequent events. Such real-time delay propagation is outside the scope of the present tactical planning model.

To address situation (b) in Fig. 8 regarding the space constraint, we compute the number of available seats based on the maximum train capacity C and the number of movements in and out at previous stations along the route of a train, as

$$C_2 = C - \sum_{s'} \left(B_j^{s'} + A_j^{s'} \right),$$

where s' denotes stations visited prior to station s. Thus, we formulate the passenger-loading capacity based on the space constraint as

$$\begin{aligned} C^s_{j,\mathrm{space}} &= S^s_j \cdot C, \quad \forall (s = 1 \wedge j \in J^{\mathrm{up}}) \ \vee \ (s = s_t \wedge j \in J^{\mathrm{dn}}), \\ C^s_{j,\mathrm{space}} &= S^s_j \cdot C_2, \qquad \forall j \in J^{\mathrm{up}} \cup J^{\mathrm{dn}},\ \forall s \notin \{1, s_t\}. \end{aligned} \tag{18}$$

Based on both space and time constraints discussed above, we compute the overall passenger-loading capacity as

$$C_j^s = \min\left\{C_{j,\text{time}}^s,\ C_{j,\text{space}}^s\right\}. \tag{19}$$

We determine the number of passengers boarding the train allocated to journey j at station s and travelling to station s' by

$$B_j^{(s,s')} = S_j^s \cdot S_j^{s'} \cdot \sum_{t=t'}^{t_j} Q_s^{s'}(t), \qquad \forall j \in J^{\text{up}} \cup J^{\text{dn}},\ \forall s, s' \in S. \tag{20}$$

where t_j represents the instant at which the number of passengers getting on the train reaches $B_j^s = C_j^s$, and t' represents the instant at which the first passenger arrives at station s intending to board journey j. For the variable t_j, we differentiate between the first journey of the day and subsequent journeys as

$$t_j = \begin{cases} t_{\text{earliest}}, & \text{if } j = 1, \\ t_{j-1}, & \text{if } j \geq 2. \end{cases}$$

Based on the variables discussed above, passengers' movements from each scheduled journey at every station can be formulated as

$$B_j^s = \sum_{s'} B_j^{s,s'}, \qquad \forall j \in J^{\text{up}} \cup J^{\text{dn}},\ \forall s \in S, \tag{21}$$

$$A_j^s = \sum_{s'} B_j^{s',s}, \qquad \forall j \in J^{\text{up}} \cup J^{\text{dn}},\ \forall s \in S. \tag{22}$$

4.4 Objective Functions

The model considers three objectives: maximizing the proportion of served passengers, minimizing operating cost, and minimizing total passenger waiting cost. Let Z_1 , Z_2, and Z_3 denote these three objectives, respectively. The multi-objective formulation is defined as follows:

$$Z_1 = \max\left(\frac{\sum_j \sum_s B_j^s}{\sum_t \sum_s \sum_{s'} Q_s^{s'}(t)}\right), \tag{23}$$

$$Z_2 = \min\left(u_c \cdot O_c\right), \tag{24}$$

$$Z_3 = \min\left(p_v \cdot W_t\right). \tag{25}$$

The first objective is formulated as a served-passenger ratio rather than as the absolute number of served passengers. This design is intentional. When total demand is fixed,

maximizing the numerator $\sum_j \sum_s B_j^s$ is mathematically equivalent to maximizing the ratio in Eq. 23. However, the ratio form is more informative when comparing different test instances, demand scenarios, or time-discretisation settings, because it expresses service coverage relative to total demand. It can also be interpreted equivalently as minimizing the stranded-passenger ratio, i.e., $1 - Z_1$.

The three objectives have different physical units and numerical magnitudes: Z_1 is dimensionless, Z_2 is expressed in monetary operating cost, and Z_3 is expressed in passenger waiting-time cost. In the proposed Pareto-based framework, the objectives are not combined through a weighted sum in the main optimization model; therefore, differences in magnitude do not bias dominance relations. Nevertheless, for diversity preservation and performance evaluation, objective values are normalized when computing crowding-distance and related quality indicators. Specifically, for each objective *m*, the normalized value of solution Λ is computed as

$$\widehat{Z}_m(\Lambda) = \frac{Z_m(\Lambda) - Z_m^{\min}}{Z_m^{\max} - Z_m^{\min}}, \qquad m \in \{1, 2, 3\}, \tag{26}$$

where $Z_m^{\min}$ and $Z_m^{\max}$ denote the minimum and maximum values of objective *m* among the candidate solutions being compared. This normalization is used only for comparison and diversity metrics; the reported objective values remain in their original units for interpretability.

In practice, a high number of journeys can improve passenger comfort, but it comes at the cost of high operating expenses. One of the main criteria for passenger satisfaction is how effectively the schedule transports the daily travel demand. This can be influenced by managing the number of scheduled journeys and their respective transit paths. On the other hand, the accumulated running time of trains is a major driver of the total operating cost of the rail line in terms of energy and maintenance of infrastructure and equipment.

We first compute the total operating cost of trains as

$$\begin{aligned} O_c \;=\; \sum_{j} \sum_{j'} \sum_{s} \Big(& I_j^s \cdot \left(E_j^{\mathrm{arr},s'} - E_j^{\mathrm{dep},s}\right) + S_j^s \cdot \left(E_j^{\mathrm{dep},s} - E_j^{\mathrm{arr},s}\right) \\ & + L_{j,j'}^s \cdot \left(E_{j'}^{\mathrm{dep},s} - E_j^{\mathrm{arr},s}\right) + t_{\mathrm{out}} \cdot C_j^s + t_{\mathrm{in}} \cdot R_j^s \Big), \end{aligned} \tag{27}$$

where

$$s' = \begin{cases} s+1, & \forall j \in J^{\mathrm{up}}, \\ s-1, & \forall j \in J^{\mathrm{dn}}. \end{cases}$$

An important aspect of schedule efficiency from the passenger perspective is the total waiting time at stations (Lei et al. 2022). We formulate the total waiting time of passengers as

$$W_t = \sum_j \sum_s (U_j^s - t) \cdot B_j^s + \sum_{j'} \sum_s (D_{j'}^s - t) \cdot B_{j'}^s, \tag{28}$$

where t represents the instant at which a group of passengers arrived at station s.

4.5 Model Properties and Computational Complexity

The proposed formulation combines binary journey-selection variables, route and stop decisions, journey-linking variables, continuous event-time variables, and integer passenger-flow variables. The resulting model is nonlinear because several constraints contain products of decision variables, minimum operators, and integer-rounding operations. Table 4 summarizes the main nonlinear components of the formulation and identifies their mathematical forms.

Most of these nonlinearities can be linearized using standard auxiliary-variable and big-M reformulations when exact solution of small instances is required. For example, products between binary variables and continuous event-time variables can be replaced by auxiliary variables with upper and lower big-M bounds. The minimum relation in Eq. 19 can be represented by linear inequalities enforcing $C_j^s \leq C_{j,\text{time}}^s$ and $C_j^s \leq C_{j,\text{space}}^s$, together with objective- or assignment-driven constraints that prevent artificial underestimation. The floor operator in Eq. 17 is implemented by defining the corresponding capacity variable as an integer variable bounded by the right-hand-side expression.

The computational difficulty of the model is therefore not caused by a single nonlinear term, but by the simultaneous presence of binary journey-selection, stopping-pattern, routing, train-circulation, event-time, and passenger-flow decisions. The number of potential journeys grows as the time unit u becomes smaller, while each additional journey generates station-level event-time variables, stopping decisions, linking decisions, and passenger-assignment variables. This creates a rapidly expanding search space and motivates the use of the tailored NSGA-II procedure described in Section 5.

The number of generated variables and constraints depends on the specific instance dimensions, including the number of stations, candidate journey time slots, depots, turnaround stations, and origin–destination station pairs. Therefore, the instance-specific model-size estimates are reported in Section 6.2, after the numerical case-study parameters have been introduced.

5 Solution Approach

Solving the proposed MINLP model is computationally demanding due to its combinatorial nature, nonlinearity, and multi-objective structure. Exact methods could be applied only to highly simplified or small-scale instances and would typically

Table 4 Main nonlinear components of the proposed formulation

Model component	Mathematical form	Source of nonlinearity
Headway activation constraints	$I_j^s I_{j'}^s$, in Constraints Eqs. 1–2	Bilinear product of binary journey-selection variables.
Arrival-time propagation	$I_j^{s''} E_j^{\mathrm{dep},s}$ and $P_j^{s''} E_j^{\mathrm{dep},s}$ in Constraint Eq. 6	Bilinear products between binary routing/selection variables and continuous event-time variables.
Intermediate departure propagation	$I_j^{s'} E_j^{\mathrm{arr},s}$ and $I_j^{s'} S_j^s$ in Constraint Eq. 7	Bilinear products involving binary selection/stopping variables and event-time variables.
Short-turn linking	$\left(1 - \sum_{s'} P_j^{s'}\right) I_{j'}^s$, in Constraint Eq. 13	Bilinear product of binary route and journey-selection variables.
Turnaround occupancy conditions	$E_{j_1}^{\mathrm{arr},s} L_{j_1,j_2}^s$ and $E_{j_2}^{\mathrm{dep},s} L_{j_1,j_2}^s$ in Constraint Eq. 16	Bilinear products between continuous event-time variables and binary linking variables.
Dwell-time-based passenger throughput	$\left\lfloor \frac{\mathrm{DT}^s - t_r}{t_m} - A_j^s \right\rfloor$ in Constraint Eq. 17	Integer-rounding operator used to represent the number of passenger movements that can be completed within planned dwell time.
Effective available capacity	$C_j^s = \min\{C_{j,\mathrm{time}}^s, C_{j,\mathrm{space}}^s\}$ in Constraint Eq. 19	Minimum operator over two capacity bounds.
Passenger assignment	$S_j^s S_j^{s'} \sum_{t=t'}^{t_j} Q_s^{s'}(t)$ in Constraint Eq. 20	Bilinear product of binary stopping variables and variable-dependent summation limit.
Operating-cost expression	$I_j^s (E_j^{\mathrm{arr},s'} - E_j^{\mathrm{dep},s})$, $S_j^s (E_j^{\mathrm{dep},s} - E_j^{\mathrm{arr},s})$, and $L_{j,j'}^s (E_{j'}^{\mathrm{dep},s} - E_j^{\mathrm{arr},s})$ in Eq. 27	Bilinear products between binary service/circulation variables and continuous event-time variables.
Waiting-time expression	$(U_j^s - t) B_j^s$ and $(D_{j'}^s - t) B_{j'}^s$, in Eq. 28	Product of waiting-time terms and passenger-flow variables.

return a single weighted-sum solution rather than a well-distributed Pareto front. For operational planning, however, it is important to visualise the trade-offs between passenger service level and operating cost. For these reasons, we adopt a Non-dominated Sorting Genetic Algorithm II (NSGA-II) framework (Deb et al. 2002), which has proven effective for multi-objective, large-scale transport planning problems and can be adapted to exploit problem-specific structure.

5.1 Chromosome Representation and Encoding

In the proposed framework, each feasible set of operated journeys satisfying constraints Eqs. 1–16 constitutes a candidate solution. A solution is defined by the tuple

$$\Lambda = \{J, P, S, TT, AP\},$$

where J denotes the set of selected journeys (I_j^s), P the associated routing decisions (P_j^s), S the stopping pattern (S_j^s), TT the resulting timetable (events $E_j^{\mathrm{dep},s}$ and $E_j^{\mathrm{arr},s}$), and AP the train allocation plan (O_j^s, R_j^s, and $L_{j,j'}^s$). Among these components, the NSGA-II operator acts primarily on the chromosomes $\{J, P, S\}$, where each gene corresponds to the decision variables I_j^s, P_j^s, and S_j^s, as illustrated in Fig. 9.

Given a feasible chromosome satisfying constraints Eqs. 2–5, the algorithm generates the timetable, train allocation plan, and passenger flow assignment. Departure times from terminal stations $s \in \{1, s_t\}$ for the first departing journeys ($j = 1$) are initialised using constraint Eq. 5. For subsequent journeys ($j > 1$), departure events are computed using constraint Eq. 6. If a feasible link exists between journey j and an opposite journey j' at the terminal station (i.e., constraints Eqs. 17–18 hold), the algorithm assigns $L_{j',j}^s = 1$, $O_j^s = 0$, and $R_{j'}^s = 0$. Otherwise, journey j must originate from the depot ($O_j^s = 1$), and the previous journey j' must return to the depot ($R_{j'}^s = 1$), ensuring satisfaction of constraints Eqs. 14–16.

At intermediate turnaround stations $s \in \{s_{i1}, s_{i2}\}$, if a journey was already initiated from a terminal station ($E_j^{\mathrm{dep},1} \neq 0 \ \vee \ E_j^{\mathrm{dep},s_t} \neq 0$), the departure event at the turnaround station is computed according to constraint Eq. 8. If this event violates constraint Eq. 9, it is adjusted; if it violates the daily time window constraints Eqs. 10–11, the journey is cancelled (i.e., all associated departure and arrival events are set to zero). If the journey did not originate from a terminal station, it may instead originate from the turnaround station provided a feasible link with an opposite journey satisfies constraints Eqs. 16–18.

For all remaining intermediate stations, departure events follow constraint Eq. 7 and must satisfy constraints Eqs. 8–11. Arrival events are computed using constraint Eq. 7 and adjusted if necessary to satisfy constraints Eqs. 10–11, or cancelled if they violate Eq. 12. If the resulting allocation plan exceeds the maximum number of available trains, the solution is declared infeasible. Finally, passenger assignment is performed based on the passenger-loading capacity constraints Eqs. 19–21 and the

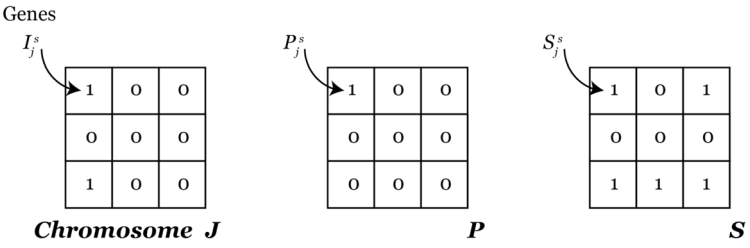


Fig. 9 The genetic concept for journey scheduling

boarding and alighting relations Eq. 22–28. The full pseudo-code of the procedure is presented in Algorithm 1.

Input: Journey selection I_j^s, routing P_j^s, stopping pattern S_j^s
Output: Timetable $E_j^{\text{dep},s}$, $E_j^{\text{arr},s}$, train allocation, passenger flows
for each journey j **do**
    Compute departure times using terminal constraint Eq. 5 and intermediate constraint Eq. 7
    Compute arrival times using propagation Eq. 6
    Enforce headway constraints Eqs. 1–2
    Enforce safety and daily window constraints Eqs. 8–11
    Update linking decisions using turnaround constraints Eqs. 12–16
    Compute loading capacities and passenger flows (Eqs. 17–22)
end for

Algorithm 1 Generation of timetable, train allocation plan, and traffic assignment.

5.2 Generating Solutions Based on Travel Demands

In the first generation, a set of P solutions is generated. The standard way to implement an evolutionary algorithm is to generate solutions randomly. However, the goal of the present algorithm is to improve search efficiency; therefore, we introduce a selective initialisation procedure. For instance, high travel demand requires higher service frequency and vice versa. We introduce a time-difference parameter to influence the selection of journeys while maintaining feasibility via the headway constraints Eqs. 1 and 2:

$$h^i = \begin{cases} \text{rand}\,[t_{\min}, t_{\max}], & \text{outside peak hours,} \\ \text{rand}\,[t_{\min}, 1.5 \cdot t_{\min}], & \text{during peak hours.} \end{cases} \tag{29}$$

Overall, Algorithm 2 generates the first population, noting that each solution must be evaluated based on Algorithm 1.

Input: $t_{\min}$, $t_{\max}$, peak-hour intervals, sets of time intervals i and journeys j
for each time interval i **do**
    if i is within peak hours **then**
        $h^i \leftarrow \text{rand}[\, t_{\min},\ 1.5 \cdot t_{\min}\,]$
    else
        $h^i \leftarrow \text{rand}[\, t_{\min},\ t_{\max}\,]$
    end if
end for
for each journey j **do**
    $I_j^1 \leftarrow \text{rand}\{0, 1\}$
    if $I_j^1 = 1$ **then**
        $P_j^1 \leftarrow \text{rand}\{0, 1\}$
        $S_j^1 \leftarrow \text{rand}\{0, 1\}$
        Enforce feasibility based on constraints Eqs. 4 and 5
    end if
end for

Algorithm 2 Initialization of headways and first-journey decisions.

5.3 Non-dominated Sorting

For the subsequent generations, offspring solutions are generated through selection, crossover, and mutation operators. Selection relies on the concept of dominance depth: a solution Λ_1 is said to dominate a solution Λ_2 (denoted $\Lambda_1 \prec \Lambda_2$) if it is no worse in all objectives and strictly better in at least one of them. Formally,

$$\Lambda_1 \prec \Lambda_2 \iff \begin{cases} \mathrm{Obj}_k(\Lambda_1) \leq \mathrm{Obj}_k(\Lambda_2), & k = 1, 2, 3, \\ \mathrm{Obj}_k(\Lambda_1) < \mathrm{Obj}_k(\Lambda_2) & \text{for at least one } k \in \{1, 2, 3\}. \end{cases} \tag{30}$$

As solutions are compared to each other, they are partitioned into non-dominated fronts according to their dominance depth, as illustrated in Fig. 10. For example, a solution Λ_1 in the first front is not dominated by any solution from lower-ranked fronts (e.g., ranks 2, 3, or 4). Similarly, a solution Λ_2 in the second front is not dominated by any solution in ranks 3 and 4, and so on.

5.4 Crowding Distance

The crowding distance measures the relative isolation of a solution within its non-dominated rank, as illustrated in Fig. 11. This metric promotes diversity by favouring solutions that lie in less crowded regions of the objective space. For each objective *m*, the crowding distance of solution Λ is computed using

$$d_{\Lambda}^{(m)} = \frac{\left|\mathrm{Obj}_{\Lambda+1}^{(m)} - \mathrm{Obj}_{\Lambda-1}^{(m)}\right|}{\mathrm{Obj}_{\max}^{(m)} - \mathrm{Obj}_{\min}^{(m)}}, \tag{31}$$

where $\Lambda \pm 1$ denote the immediate neighbors of Λ within the same rank, and $\mathrm{Obj}_{\max}^{(m)}$ and $\mathrm{Obj}_{\min}^{(m)}$ represent, respectively, the maximum and minimum values of objective *m* within that rank. As shown in Fig. 11, solution Λ_i has a larger crowding distance than solution Λ_j, indicating that it lies in a less dense region of the Pareto front.

5.5 Crowded Binary Non-dominated Selection

Selection for crossover relies on non-dominated sorting and crowding distance. Given two candidate solutions Λ_1 and Λ_2, the selection procedure follows three principles:

- If Λ_1 and Λ_2 belong to different ranks, the solution in the higher (better) rank is selected.
- If Λ_1 and Λ_2 lie in the same rank, the solution with the larger crowding distance is selected.
- If the two solutions share both the same rank and the same crowding distance, one of them is chosen uniformly at random.

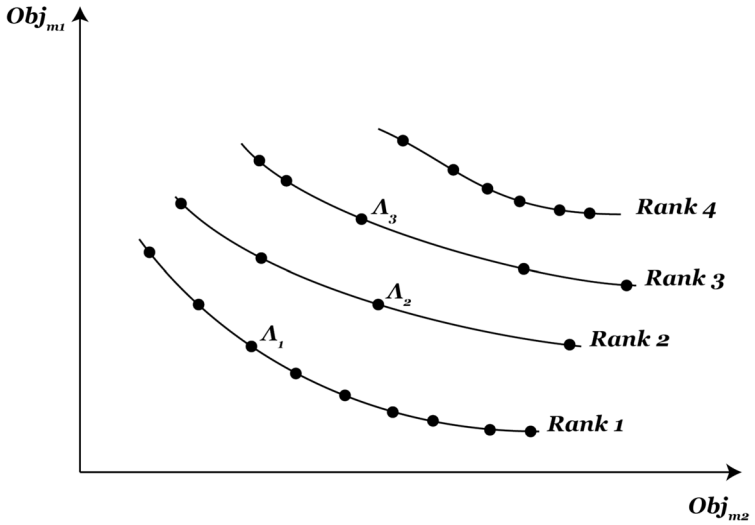


Fig. 10 Solutions ranking based on non-dominated sorting

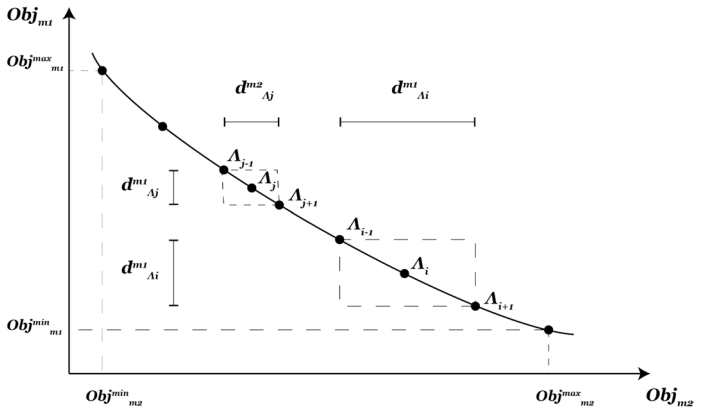


Fig. 11 Principle of crowding distance within a rank of solutions

Because all chromosome components (J, P, S) are binary, crossover is performed by selecting a random crossover point within each chromosome and exchanging the corresponding subsequences between two parents. Mutation is applied to each gene with a small probability $P(\text{rand} < M_r)$, where M_r denotes the mutation rate. Survivor selection is then performed using the combination of non-dominated sorting and crowding distance ordering, as illustrated in Fig. 12.

The evolutionary process continues until the predefined number of generations G is reached, at which point the algorithm terminates and returns the approximate Pareto front. The general procedure is summarised in Algorithm 3.

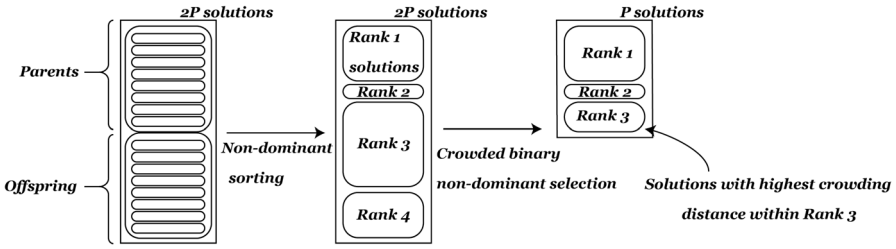


Fig. 12 Survival of the best

1: **if** $g = 1$ **then**
2: Generate an initial population using Algorithms 2 and 1
3: Evaluate all solutions and assign ranks and crowding distances
4: **else**
5: Select parent solutions via crowded binary tournament selection
6: Apply crossover and mutation operators to generate offspring
7: Evaluate all offspring solutions (Algorithm 1)
8: Merge parent and offspring populations and select survivors
9: **end if**
10: **if** $g < G$ **then**
11: $g \leftarrow g + 1$
12: **go to** step 2
13: **else**
14: **Terminate** and return the final Pareto front
15: **end if**

Algorithm 3 NSGA-II evolutionary framework.

5.6 Role of the MINLP Formulation and Originality of the Algorithmic Design

Although NSGA-II is a well-established metaheuristic, its effectiveness in this study relies on a tight coupling with the underlying MINLP formulation. The model plays two key roles.

First, it provides a *structured encoding* of solutions: chromosomes are defined directly in terms of model variables (I_j^s, P_j^s, S_j^s), and the remaining components (timetable, circulation, and passenger flows) are derived by enforcing the constraints of Section 4. This leads to feasibility-preserving operators that automatically satisfy headway, turnaround, and operating-window constraints, instead of relying on ad hoc penalty functions or external repair procedures.

Second, the MINLP formulation defines physically interpretable objective functions and passenger-loading capacity constraints that guide the evolutionary search. The evaluation of each chromosome is not a black-box simulation, but a structured procedure (Algorithm 1) that systematically applies the model equations. This ensures consistency between the algorithmic search and the underlying operational rules of the RTS.

The originality of the solution approach therefore lies in:

- a problem-specific chromosome design that jointly captures journey selection, routing, and stopping patterns;

- an evaluation mechanism that embeds timetable generation, train circulation planning, and time-dependent passenger assignment within each fitness computation; and
- a demand-driven initialisation strategy and feasibility-preserving operators tailored to the integrated RTS planning problem.

These elements go beyond a direct application of NSGA-II and are essential to handle the high dimensionality and tight coupling of decisions in the proposed integrated planning framework.

6 Numerical Experiments

Because the proposed formulation combines binary journey–selection variables, time-dependent passenger flows, and circulation constraints, exact optimization is only practical on extremely small toy instances. For such reduced configurations, commercial MINLP solvers are able to enumerate the optimal solution, and the structure of the resulting schedules confirms that our encoding and evaluation mechanisms reproduce the same sequencing, stopping, and circulation decisions as the exact model. Although full-scale exact optimization is computationally infeasible due to memory limits and branch-and-bound growth, these small-instance experiments provide qualitative evidence that the proposed NSGA-II procedure is consistent with the underlying mathematical formulation and behaves as a faithful approximation method whenever exact solutions are obtainable.

Given these tractability limitations, the numerical experiments focus on evaluating the solution quality of the proposed NSGA-II procedure through controlled comparisons with established single- and multi-objective heuristics, as well as through detailed inspections of the resulting schedules. Next, we examine the operational impact of the proposed flexible routing and stopping strategies by comparing them with conventional all-stop, skip-stop, and short-turn patterns. A sensitivity analysis is carried out to investigate the influence of key parameters such as the time-scale discretisation and mutation rate on solution quality, diversity, and computational burden. Finally, we provide a detailed inspection of selected solutions, including timetables and train circulation plans for both a small two-hour window and a full-day operating scenario, in order to highlight the practical implications of the model from an operator's perspective.

6.1 The Tractability of Exact Optimization

The proposed formulation combines binary journey-selection variables, routing and stopping decisions, train–journey linking variables, and nonlinear passenger-flow constraints. Even when the nonlinear components are relaxed, the resulting model contains a very large number of binary variables and tightly coupled temporal–spatial constraints. Preliminary exploratory runs with a commercial MINLP solver on reduced configurations led to rapid memory consumption and extremely slow progress within the branch-and-bound tree, indicating that exact optimization of realistic,

full-day scenarios is computationally prohibitive. As a result, exact methods can be applied only to highly simplified or small instances, whereas full-scale operation plans require heuristic or hybrid approaches designed to navigate the combinatorial structure of the problem efficiently.

To ensure that the proposed NSGA-II procedure is structurally consistent with the underlying MINLP formulation, we performed additional checks on highly simplified reduced configurations for which exact enumeration is still feasible. On these small instances, the sequencing of departures, stopping decisions, and circulation links generated by the metaheuristic matched the structure of the solutions returned by the exact model. Although these reduced tests cannot be extended to full-day instances due to prohibitive memory and time requirements, they confirm that the evaluation and encoding mechanisms reproduce the intended logic of the mathematical formulation and behave consistently whenever exact enumeration is computationally accessible.

6.2 Experimental Parameter Setting and Model-size Estimates

The section of Line 1 of the Xi'an metro system considered in the numerical experiments consists of 19 stations. The up-way direction operates from Houwei Zhai station, denoted as station 1, to Fangzhi Cheng station, denoted as station $s_t = 19$. Two intermediate turnaround stations are considered: Hancheng Lu ($s_{i1} = 5$) and Wanshou Lu ($s_{i2} = 15$), as illustrated in Fig. 13. The daily operation period is set from $t_{\text{earliest}} = 6{:}00$ to $t_{\text{latest}} = 00{:}00$. Passenger travel demand is obtained from real-world smart-card data for a single day, aggregated on an hourly basis. Time intervals 08:00–09:00 and 18:00–19:00 are treated as peak-demand periods.

For safety reasons, the minimum allowed time difference between consecutive trains is fixed at $t_{\min} = 3$ minutes, while the maximum time difference between journeys is set at $t_{\max} = 10$ minutes. For scalability, the time unit parameter is initialised as $u = 3$ minutes. All numerical experiments are implemented in MATLAB R2022b and executed on a machine equipped with an Intel® i5–6300HQ @ 2.30 GHz CPU and 8 GB RAM running Windows 10.

Based on these instance dimensions, we analytically estimate the number of variables and constraints generated by the formulation. These counts are tied to the Xi'an Line 1 case study and are therefore reported here rather than in the general model-formulation section.

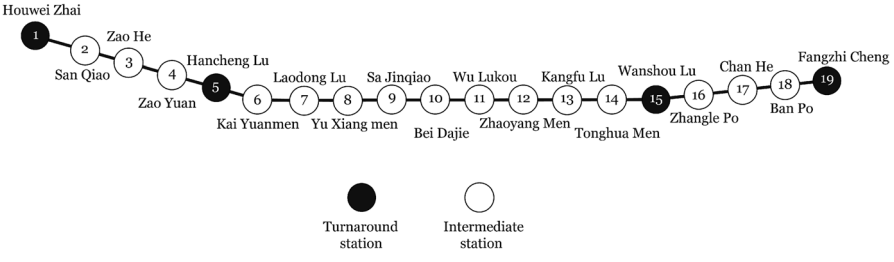


Fig. 13 Layout of Line 1 from the Xi'an metro system

Table 5 Estimated model size of the main test instances

Instance	Stations $\lvert S\rvert$	Depots	Candidate slots per direction	Estimated variables	Estimated constraints
Small-scale instance	19	2	40	46,560	196,800
Full-day instance	19	1	360	878,400	12,139,200

The counts in Table 5 are computed from the variable and constraint families listed in Section 4. Let n_J denote the number of candidate journey time slots per direction; thus, $|J^{\mathrm{up}}| = |J^{\mathrm{dn}}| = n_J$. The main variable families are counted as follows: I_j^s and P_j^s each contain $4n_J$ binary variables; S_j^s, $E_j^{\mathrm{dep},s}$, and $E_j^{\mathrm{arr},s}$ are station-level variables defined over both travel directions and therefore each contain $19 \times 2n_J$ variables; O_j^s and R_j^s each contain $2 \times 2n_J$ variables in the small-scale double-depot instance and $1 \times 2n_J$ variables in the full-day single-depot instance; $L_{j,j'}^s$ contains $4n_J^2$ binary variables because train-linking decisions may occur at the two terminals and the two intermediate turnaround stations; $C_{j,\mathrm{time}}^s$, $C_{j,\mathrm{space}}^s$, C_j^s, B_j^s, and A_j^s each contain $19 \times 2n_J$ integer variables; and $B_j^{s,s'}$ contains $19 \times 18 \times 2n_J$ integer variables. The constraint counts are obtained analogously by expanding Constraints Eqs. 1–22 over their corresponding station, direction, journey-pair, turnaround-station, depot, and origin–destination index sets.

Regarding train circulation, depot entry and exit movements each require a minimum of $t_{\mathrm{in}} = t_{\mathrm{out}} = 3$ minutes, and the minimum turnaround time between two linked journeys is set at $t_{\mathrm{turn}} = 2$ minutes. The unit operating cost of trains is fixed at $u_c = 200$ yuan/min, while the passenger time value is set to $p_v = 15$ yuan/min. Trains have a maximum capacity of $C = 600$ seats. Station dwell times are provided in Appendix A, and section running times for all interstation segments are provided in Appendix B.

6.3 Performance Analysis with Single-objective Benchmark Algorithms

In the first experiment, we evaluate the performance of the improved NSGA-II algorithm over a full-day schedule and compare it against two benchmark single-objective heuristics: Simulated Annealing (SAA) (Kirkpatrick et al. 1983 and Tabu Search (TS) (Glover 1986). Since these algorithms are designed for single-objective optimization, we adapt the model by aggregating the three objectives into a single scalar objective:

$$\min \mathrm{Obj} = -Z_1 + Z_2 + Z_3. \tag{32}$$

Because NSGA-II yields a Pareto front rather than a single optimum, we compare the best compromise solution on the Pareto front with the optimal solutions returned by SAA and TS. Table 6 summarises the results.

Table 6 Comparison of computation time and objective values for benchmark algorithms

Algorithm	Computation time (s)	Z_1 (%)	Z_2 ($\times 10^6$)	Z_3 ($\times 10^6$)
Improved NSGA-II	26400	100	125.111	238.516
SAA	26400	89	132.816	275.824
TS	26400	84	156.906	318.139

For the SAA, the initial temperature is set to $\mathrm{temp}_{\mathrm{ini}} = 100\,000$ and lowered until $\mathrm{temp}_{\mathrm{min}} = 900$ in decrements of 1000 per cooling cycle, with 40 iterations per cycle. For the TS algorithm, the Tabu tenure is set to $\mathrm{Tabu}_{\mathrm{tenure}} = 10$, with a maximum of 50 000 iterations. All parameters were calibrated to ensure that SAA and TS use approximately the same computation time as NSGA-II with 200 generations and a population size of 200.

The improved NSGA-II outperforms the benchmark algorithms across all objectives. Compared to SAA, it achieves gaps of 11.20%, 6.16%, and 15.64% for Z_1, Z_2, and Z_3, respectively. Relative to TS, the gaps increase to 5.69%, 25.41%, and 33.38%. Moreover, the Pareto front of the improved NSGA-II contains 105 non-dominated solutions. The best compromise solutions obtained by SAA and TS are dominated by at least one solution from the improved NSGA-II Pareto front, whereas neither benchmark solution dominates any point of the improved NSGA-II front.

6.4 Performance Analysis with Multi-objective Benchmark Algorithms

In this analysis, we compare the performance of the improved NSGA-II with several established multi-objective heuristics, including the standard Multi-Objective Tabu Search (MOTS) (Hansen 1997), the Multi-Objective Simulated Annealing Algorithm (MOSAA) (Czyżżak and Jaszkiewicz 1998), and the classical NSGA-II (Deb et al. 2002). For a fair comparison, the standard NSGA-II generates random initial populations, unlike the improved version, but uses identical parameter settings. For MOTS, we adopt the same Tabu tenure setting as in the previous experiment, $\mathrm{Tabu}_{\mathrm{tenure}} = 10$. For MOSAA, we match the temperature schedule and cycle structure used in the SAA benchmark. All algorithms are allocated a sufficiently large number of iterations to ensure that they run for the same duration as the improved NSGA-II, thereby allowing all methods to reach the same computational time limit. Figures 14 and 15 show the solutions obtained at the final generation for each algorithm.

After 7 hours and 20 minutes of computation time, MOTS, standard NSGA-II, and the improved NSGA-II all exhibit strong convergence behavior across objectives Z_1, Z_2, and Z_3. By contrast, MOSAA lags behind, as highlighted in Fig. 15, where the two-dimensional projections reveal a clear gap in convergence performance. A direct comparison of the resulting solution sets provides further insight into their dominance depth. The improved NSGA-II produces the largest set of non-dominated solutions, with 131 Pareto-optimal points. MOTS follows with 123 non-dominated solutions, while the standard NSGA-II returns 83 such solutions. Finally, all solutions produced by MOSAA are dominated by those obtained from the other algorithms (Fig. 16).

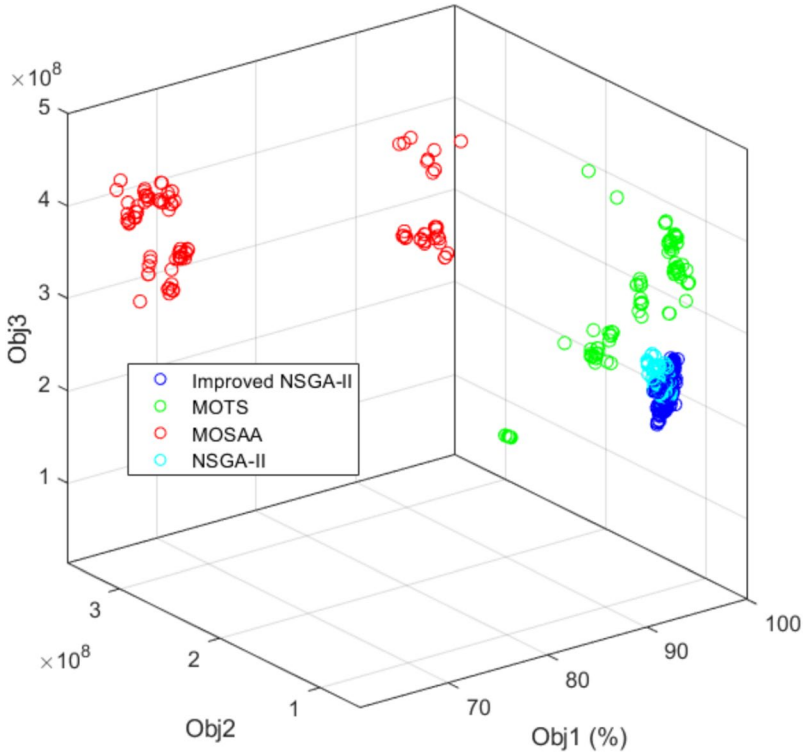


Fig. 14 3D visualisation of algorithmic performance

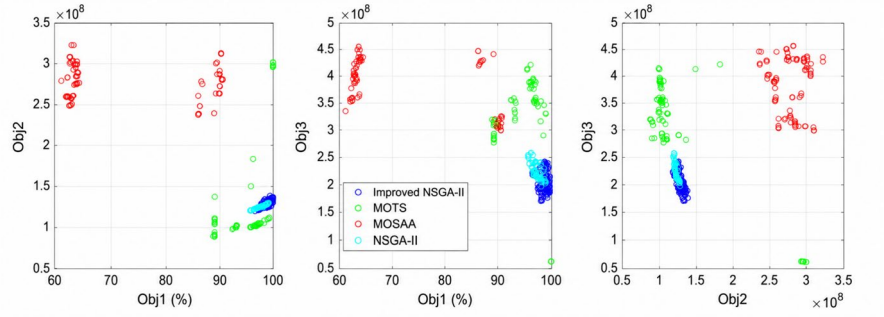


Fig. 15 Performance comparison of optimization algorithms on three objectives

6.5 Convergence and Pareto Front Analysis

To better illustrate the convergence behavior of the improved NSGA-II, Fig. 17 presents the three-dimensional Pareto fronts obtained at each generation of the algorithm. In addition, Fig. 16 provides two-dimensional views based on pairwise comparisons of the three objectives.

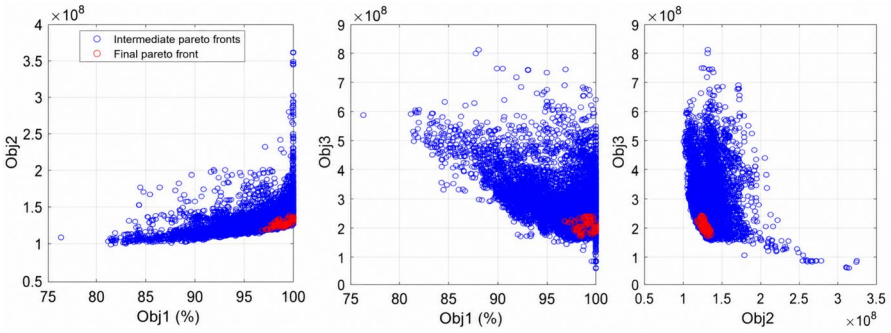


Fig. 16 2D pairwise objective comparison of Pareto front convergence

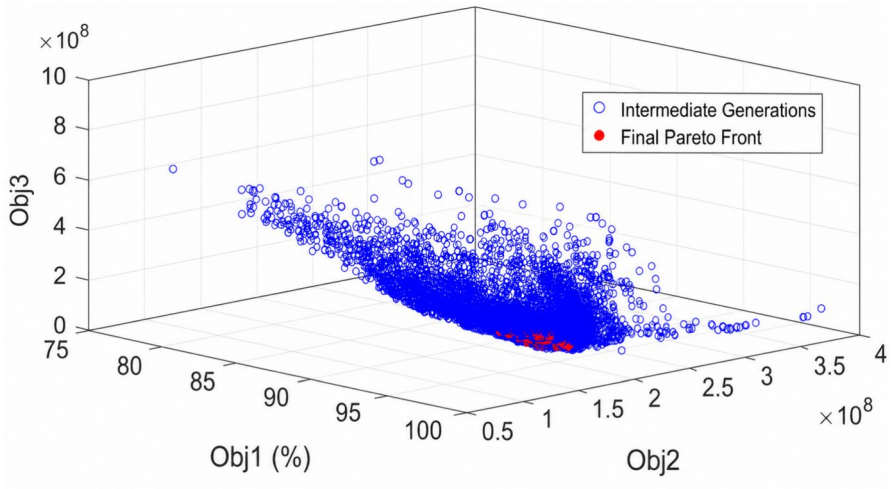


Fig. 17 Convergence of Pareto front in 3D space across 200 generations

The spacing and spread metrics offer insights into the quality of the front in terms of the distribution and diversity of solutions across the objective space. Spacing is computed using Eq. 33, while the spread is evaluated using Eq. 34.

$$S_{\text{spacing}} = \frac{1}{P}\sum_{p=1}^{P-1}\left|e_p - \bar{e}\right|, \tag{33}$$

$$S_{\Delta} = \frac{\sum_{p=1}^{P-1}\left|e_p - \bar{e}\right|}{(P-1)\left(\text{Obj}_m^{\max} - \text{Obj}_m^{\min}\right)}, \tag{34}$$

For the final generation, the spacing values obtained are $S_{\text{spacing}}^{(Z_1)} = 0.083$, $S_{\text{spacing}}^{(Z_2)} = 411651.168$, and $S_{\text{spacing}}^{(Z_3)} = 1166532.114$, each representing less than

5% of the total objective range, which is considered a good indicator of uniform spacing.

Similarly, the spread values are $S_{\Delta}^{(Z_1)} = 0.009$, $S_{\Delta}^{(Z_2)} = 0.009$, and $S_{\Delta}^{(Z_3)} = 0.038$

, all of which are close to zero, indicating that the solutions are well-distributed across the full extent of the Pareto front. The range of the extreme solutions also provides complementary information regarding the overall width of the achievable trade-offs. These observations are further examined in the sensitivity analysis to evaluate the effect of the mutation operator on the diversity and quality of the front.

6.6 Performance Analysis Compared to Skip-stop, Short-turn, and All-stop Patterns

In this analysis, we directly compare the proposed approach with flexible routing and stopping patterns against conventional all-stop, skip-stop, and short-turn strategies. These benchmark patterns can be obtained within the improved NSGA-II framework by appropriately fixing the route and stop decision variables P_j^s and S_j^s.

For a conventional all-stop pattern, both variables are fixed to $P_j^s = 1$ and $S_j^s = 1$, ensuring that all trains stop at every station and only perform turnarounds at terminal stations. For short-turning patterns, only the stop variable is fixed, $S_j^s = 1$, while routing decisions remain flexible. For skip-stop patterns, the route variable is fixed to $P_j^s = 1$, and the stop decisions S_j^s follow pre-specified skip-stop patterns, as illustrated in Fig. 1.

To compare the quality of the resulting Pareto fronts, we use the *C*-metric, defined as

$$C(f_1, f_2) = \frac{N_{f_1 > f_2}}{N_{f_2}}, \tag{35}$$

where f_1 and f_2 denote two Pareto fronts, $N_{f_1 > f_2}$ is the number of solutions in f_2 that are dominated by at least one solution in f_1, and N_{f_2} is the total number of solutions in f_2.

The results in Table 7 highlight the dominance of the flexible routing and stopping pattern over the all-stop, skip-stop, and short-turn alternatives. The proposed pattern yields a Pareto front with 113 solutions, of which 105 dominate 34 solutions from the skip-stop pattern and 46 solutions from the all-stop pattern. Moreover, 109 solutions from the flexible pattern dominate 70 solutions obtained under the traditional short-turn pattern.

6.7 Sensitivity Analysis

In this analysis, we investigate the influence of the mutation rate on the diversity and distribution of solutions within the Pareto front. In the previous experiments, the mutation rate was set to $M_r = (T + 2)\,|I|/u$. Here, we vary the mutation rate within the interval [0, 1] and evaluate its effect on the range, spacing, and spread metrics of the resulting Pareto fronts. The results reveal a clear proportional relation-

Table 7 Dominance analysis of flexible route-and-stop scheduling over alternative patterns

Alternative pattern	$C(f_1, f_2)^1$	$C(f_2, f_1)^1$
All-stop	0.657	0.071
Skip-stop	0.642	0.071
Short-turn	0.833	0.035

[1] Here, f_1 denotes the Pareto front of the flexible route-and-stop pattern, and f_2 denotes the Pareto front corresponding to the alternative transit pattern

ship between the mutation rate and the objective ranges of the solutions, indicating that higher mutation rates enlarge the explored solution space. As shown in Fig. 18, the ranges of objectives Z_1, Z_2, and Z_3 increase by 3.13%, 16.54%, and 3.51%, respectively.

Figure 19 depicts the Pareto front produced with a mutation rate of $M_r = 0.2$. The three-dimensional visualisation highlights a broad and uniform distribution of solutions, reflecting effective exploration of the search space. The solutions are not clustered, suggesting that the optimization process achieved a strong balance between exploration and exploitation. This observation is confirmed by the two-dimensional projections in Fig. 20, where solutions are well spread across each pair of objectives.

Diving deeper into the distribution of the solutions across pairs of objectives, the solutions are overall well spread between each pair, as seen in Fig. 20. Overall, the results suggest that the optimization algorithm has effectively identified a set of non-dominated solutions that offer a good balance between the three objectives.

Although mutation significantly increases the range of the Pareto front (Fig. 18), the spacing and spread indicators remain within acceptable bounds: spacing values remain below 5%, and spread values S_Δ are close to zero (Table 8), indicating well-distributed solutions.

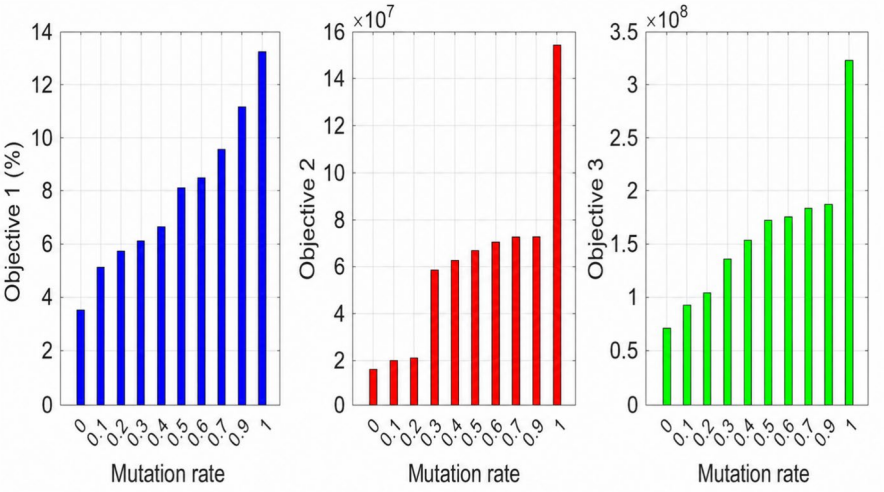


Fig. 18 Objective range analysis for different mutation rates

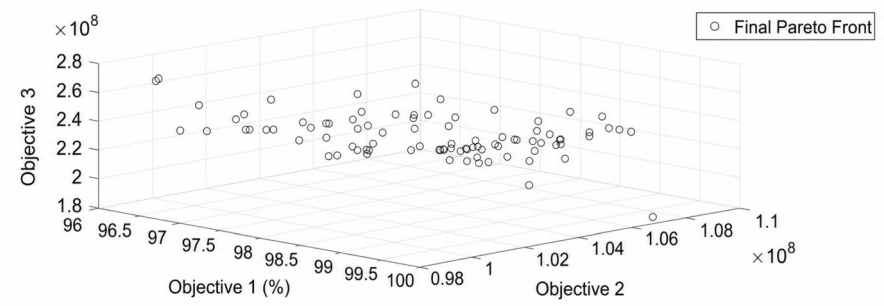


Fig. 19 3D visualisation of the Pareto front for $M_r = 0.2$ mutation rate

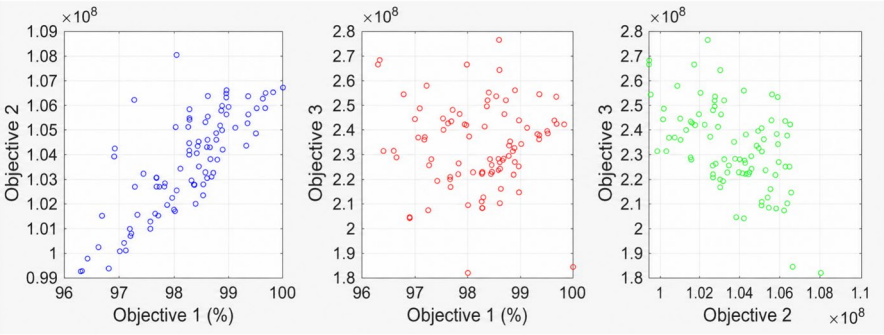


Fig. 20 2D visualisation of the Pareto front for $M_r = 0.2$ mutation rate

Table 8 Comparison of spacing and spread metrics across different mutation rates

Mutation rate	S_{spacing} (%)			S_{Δ} (%)		
	Z_1	Z_2	Z_3	Z_1	Z_2	Z_3
$M_r = 0$	0.928	0.889	3.779	0.936	0.897	3.813
$M_r = 0.1$	1.014	0.950	0.916	1.014	0.950	0.916
$M_r = 0.2$	0.847	0.809	1.034	0.847	0.809	1.034
$M_r = 0.3$	0.980	0.883	1.010	0.980	0.883	1.010
$M_r = 0.4$	0.620	0.763	0.860	1.128	0.763	0.860
$M_r = 0.5$	0.731	0.793	0.731	0.731	0.793	0.731
$M_r = 0.6$	0.760	0.854	0.737	0.760	0.854	0.737
$M_r = 0.7$	0.807	0.778	1.005	0.807	0.778	1.005
$M_r = 0.8$	0.532	0.621	0.637	0.532	0.621	0.637
$M_r = 0.9$	0.636	0.726	0.779	0.636	0.726	0.779
$M_r = 1$	1.055	1.042	0.988	1.055	1.042	0.988

We next analyze the scalability of the model by varying the time-unit parameter u. Reducing the value of u increases the temporal granularity of the model, thereby enlarging the number of potential journey combinations at the cost of higher com-

putational times. This expansion of the solution space may lead to improved Pareto fronts. To quantify these improvements, we use the hypervolume indicator, defined as

$$H_v(F) = \int I\big(\exists\, \Lambda \in F : \Lambda \prec x \prec r\big)\, dx, \tag{36}$$

where the indicator checks whether any solution Λ in the front F dominates a point x within the bounds of a reference point r.

Figure 21 illustrates the relationship between the scaling parameter u, computation time, and hypervolume. A significant improvement in hypervolume is observed when reducing u from 3 to 2.4, with only a moderate increase in computation time. However, for $u \leq 1.5$, the hypervolume drops sharply despite the substantial increase in computational cost. This degradation can be attributed to memory constraints, which restrict the allowable population size and number of generations when the model scale becomes excessively fine.

6.8 Detailed Real-world Case Study Analysis

To further analyze the performance of the flexible stopping and routing, we consider two case studies: one small case study with a 2-hour operating window and a double-depot configuration, and one full-day case study with an 18-hour operating window and a single depot. For each case study, we run a comparative analysis between the flexible scheduling approach with enhanced short-turning and stopping patterns and a regular schedule featuring an all-stop pattern with turnarounds at terminal stations only. We maintain the same model scaling at $u = 3$ minutes.

6.8.1 Case Study A: Small Case Study with Double Depot

In Case A, the operating period consists of two intervals from 6:00 to 8:00, with additional buffer time allowing trains to return to the depot. The first and last journeys depart from Houwei Zhai and Fangzhi Chen stations, respectively, at 6:00. Figure 22 presents the timetable for the regular schedule, while Fig. 23 illustrates its corre-

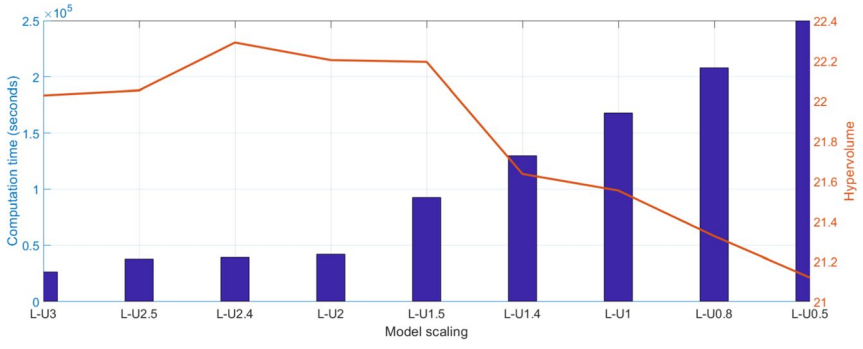


Fig. 21 Impact of the scalability parameter on computation time and the Pareto front

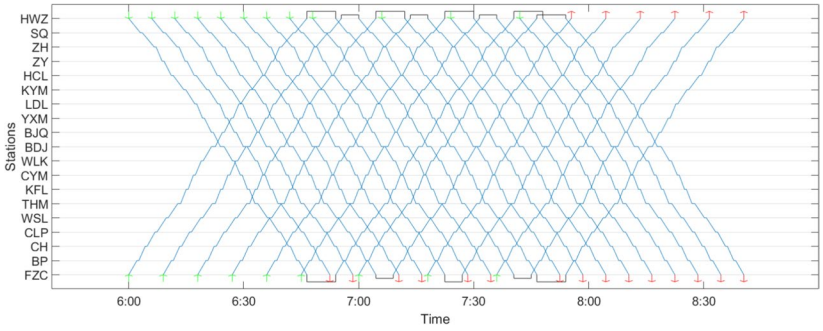


Fig. 22 Illustration of the 2-hour interval regular timetable in Case A

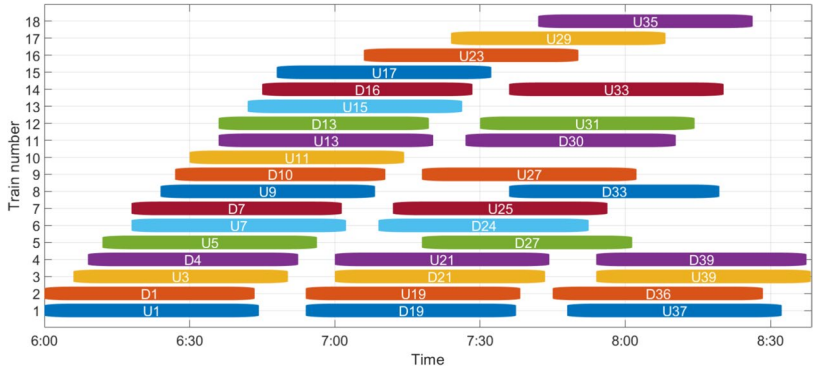


Fig. 23 Illustration of the regular train circulation plan with 34 journeys in Case A

sponding train circulation plan. The timetable and train circulation plan generated by the improved NSGA-II approach are shown in Figs. 24 and 25, where skipped stations are denoted by black circles in Fig. 24.

In the enhanced schedule, both running times and dwell times vary between journeys. Consequently, the headway is not uniform across the up-way and down-way directions. In addition, turnaround times at terminal stations differ, further contributing to variations in headways between the two directions. The enhanced schedule successfully modifies these headways while still respecting the minimum safety requirements. It also introduces a short-turn operation during the second interval at Wanshou Lu station, with a total of 14 skipped stations and 752 stops, compared to 646 stops in the regular schedule.

These adjustments increase the proportion of travelers successfully served and yield substantial improvements in multiple objectives. As shown in Table 9, the enhanced schedule reduces operating costs by 12.987% and passenger waiting time by 3.036%. The reduction in waiting time results from strategically skipping less crowded stations and employing short-turning at Wanshou Lu to better serve the most congested corridor segment. Operating costs decrease despite a 15% increase in the

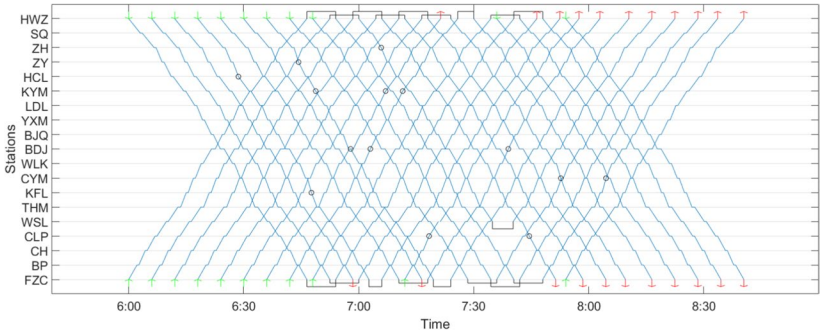


Fig. 24 Illustration of the timetable of the enhanced schedule in Case A

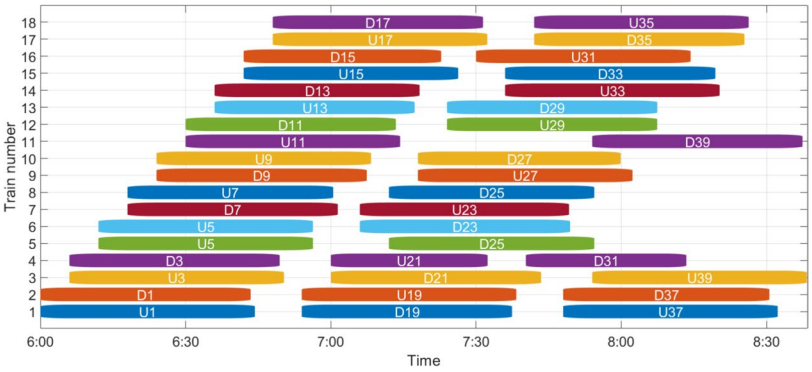


Fig. 25 Train circulation plan of the enhanced schedule with 40 journeys in Case A

Table 9 Comparison of regular and enhanced schedules for Case A

Schedule	Travelers rate (%)	Oper. costs ($\times 10^6$)	Wait. time ($\times 10^6$)	Trains	Journeys	Stops	Skips	Long route	Short route
Regular schedule	98.76	8.159	6.01	18	34	646	0	34	0
Enhanced schedule	100	7.099	5.83	18	40	752	14	39	1

number of journeys, owing to fewer station stops and shorter turnaround operations. Furthermore, the proposed plan requires only 44 depot circulations to complete 40 journeys, compared to the 42 circulations required to complete 34 journeys under the regular schedule.

In summary, the enhanced schedule allocates the same number of trains (18) as the regular schedule, while delivering superior service efficiency. Specifically, 18 trains support 40 journeys in both up-way and down-way directions, compared to only 34 journeys under the regular schedule.

6.8.2 Case Study B: Full Case Study with a Single Depot

In Case B, passenger-service departures cover the full-day operating period from 6:00 to 00:00, with an additional post-service circulation buffer allowing the last train to return to the depot before 01:00. The instance uses a single depot connected to station Houwei Zhai. The first journey departs from station Houwei Zhai at 6:00, while the last journey returns to station Houwei Zhai before 01:00. The alternative regular schedule is illustrated in Figs. 26 and 27, respectively representing the timetable and train circulation plan. The regular schedule features lower time differences during peak intervals, leading to a 100% traveller rate. Likewise, the enhanced schedule presented in Figs. 28 and 29 also achieves a 100% traveller rate.

The enhanced full-day timetable features 34 short-turns at station Wanshou Lu with a total of 77 skips and 7121 stops, compared to 7467 stops for the regular schedule. A slight variation in the time difference between journeys can be observed for departures from Fangzhi Chen station due to the time required for trains to be available. Nevertheless, the proposed approach maintains the 100% traveller rate. Additionally, we observe a 3.760% reduction in operating costs and a 32.694% decrease in the waiting time of passengers, as shown in Table 10.

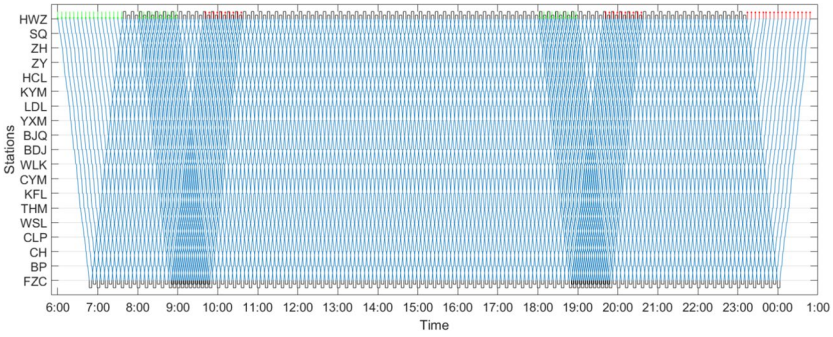


Fig. 26 Illustration of the regular full-day schedule timetable in Case B

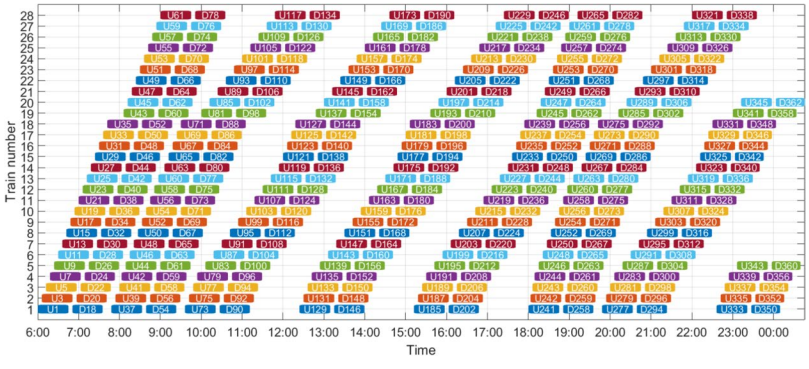


Fig. 27 Illustration of the regular train circulation plan with 393 journeys in Case B

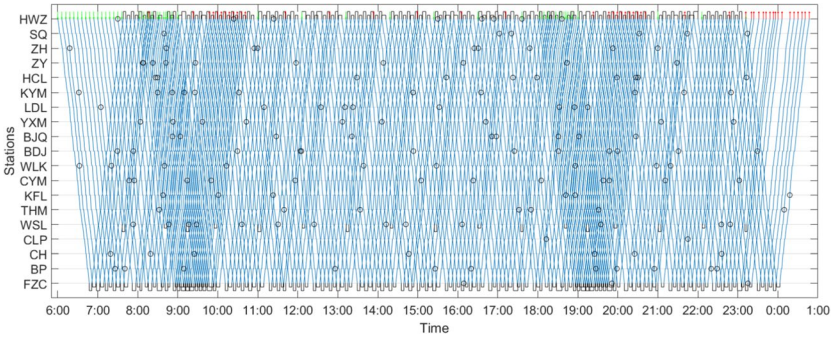


Fig. 28 Illustration of the timetable of the enhanced full-day schedule in Case B

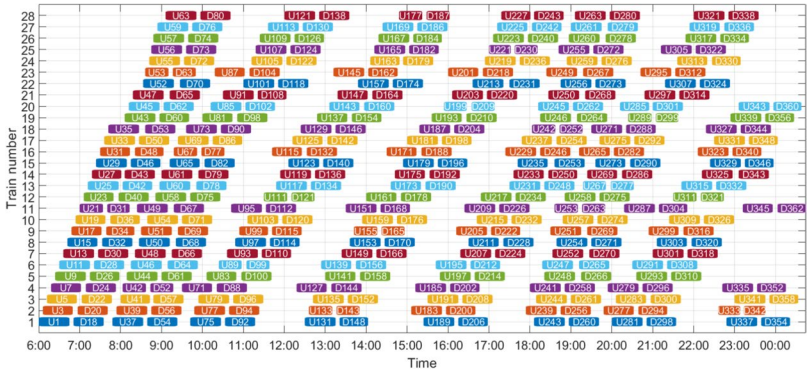


Fig. 29 Illustration of the enhanced train circulation plan with 393 journeys in Case B

Table 10 Comparison of regular and enhanced schedules for Case B

Schedule	Travelers rate (%)	Oper. costs ($\times 10^8$)	Wait. time ($\times 10^8$)	Trains	Journeys	Stops	Skips	Long route	Short route
Regular schedule	100	1.530	2.266	28	393	7467	0	393	0
Enhanced schedule	100	1.472	1.525	28	393	7121	77	352	34

In terms of train circulation planning, the enhanced schedule allocates 28 trains to a total of 393 journeys in both up-way and down-way directions, identical to the regular schedule. Overall, flexible scheduling improves the utilisation of the fleet while ensuring the same number of passengers is transported, with better waiting time and operating cost performance.

The experimental results demonstrate the superiority of the proposed integrated model over both traditional fixed-pattern strategies and classical metaheuristics. While skip-stop and short-turn plans offer moderate efficiency gains, their rigid application across all time intervals results in suboptimal load balancing and under-

utilisation during off-peak periods. In contrast, the proposed model adaptively adjusts stopping patterns and turnaround points in response to time-varying demand, leading to more balanced train occupancy and significantly reduced waiting times. The flexibility of journey configurations achieved without increasing the number of trains proves critical in matching supply with spatial demand asymmetries.

Importantly, the Pareto fronts obtained using the enhanced NSGA-II framework are not only broader but also more diverse and better structured than those produced by competing algorithms. This suggests that, beyond algorithmic tuning, it is the hybrid integration of domain-specific constraints with evolutionary search that drives performance. However, this gain in solution quality comes with increased computational cost, particularly when finer time granularity or high mutation rates are used. Future extensions should explore surrogate-based modeling or hybrid exact/metaheuristic decomposition to enhance scalability without compromising result interpretability or responsiveness.

7 Conclusion

This study developed an integrated multi-objective model for rapid transit opera tion planning that simultaneously optimises route selection, stop patterns, timetable generation, and train circulation under time-dependent demand. Using a MINLP formulation and a customised NSGA-II algorithm, the framework allows for flex ible adaptation of service patterns including skip-stop and short-turn operations based on passenger flow heterogeneity. Applied to Xi'an Metro Line 1, the proposed approach consistently outperforms conventional and heuristic-based benchmarks in terms of demand satisfaction, waiting time, and operating costs, without necessitating additional rolling stock.

In addition to its numerical performance, the study contributes in three key areas: (i) a fully integrated planning model unifying strategic and tactical decisions; (ii) an algorithmic design that embeds feasibility into the evolutionary encoding process; and (iii) a practical demonstration that flexibility in service design can yield operational gains. Across the numerical experiments, the customised NSGA-II framework yields well-distributed Pareto fronts and consistently dominates benchmark heuristics, providing high-quality approximations for full-day operational settings that lie far beyond the practical reach of exact approaches.

Nevertheless, computational costs increase under tight fleet constraints, and the model assumes deterministic travel demand and passenger behavior. Future work should explore robust formulations under uncertainty, extend the framework to include energy consumption, and evaluate its applicability in real-time rescheduling contexts with live data feeds. These extensions would enhance the model's utility for modern transit agencies facing fluctuating demand and sustainability pressures.

Appendix A Stopping Times at Stations

Table 11 Stopping time of trains at each station (minutes)

Station	Stopping time (min)
HWZ	0.5
SQ	0.5
ZH	0.5
ZY	0.75
HCL	0.75
KYM	0.75
LDL	0.75
YXM	0.75
BJQ	0.75
BDJ	0.5
WLK	0.5
CYM	0.5
KFL	0.5
THM	0.75
WSL	0.75
CLP	0.75
CH	0.75
BP	0.5
FZC	0.5

Appendix B Running Times Between Stations

Table 12 Running times between consecutive stations (minutes). For each section A–B, the "stop" value refers to a train that stops at the downstream station B, while the "skip" value refers to a train that passes station B without stopping

Section	Stop running time (min)	Skip running time (min)
HWZ – SQ	3.0	2.3
SQ – ZH	3.1	2.3
ZH – ZY	1.6	1.2
ZY – HCL	1.9	1.4
HCL – KYM	2.8	2.1
KYM – LDL	2.4	1.8
LDL – YXM	1.5	1.1
YXM – BJQ	1.8	1.4
BJQ – BDJ	1.8	1.4
BDJ – WLK	2.2	1.6
WLK – CYM	1.5	1.2
CYM – KFL	1.4	1.0
KFL – THM	1.4	1.1
THM – WSL	2.5	1.9
WSL – CLP	2.4	1.8
CLP – CH	2.1	1.5
CH – BP	1.7	1.3
BP – FZC	2.1	1.6

Author Contributions M.O. conceived the study, developed the model, conducted the numerical experiments, and drafted the manuscript. W.Z. supervised the research, contributed to the interpretation of results, and reviewed the manuscript. All authors reviewed the results and approved the final version of the manuscript.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Use of AI-assisted technologies During the preparation of this work, the authors used ChatGPT to improve the clarity of some sections. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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